

**CROSS-FUNCTIONAL TEAM PERFORMANCE:
INQUIRY, IDENTITY, AND SHARED REALITY**

by

SUSAN LEIGH JOHNSON
Design and Innovation Fellow

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CASE WESTERN RESERVE UNIVERSITY
SCHOOL OF GRADUATE STUDIES

We hereby approve the dissertation of

Susan Leigh Johnson

candidate for the degree of Doctor of Philosophy.*

Committee Chair

Richard Boland, Ph.D., Case Western Reserve University

Committee Member

Philip Cola, Ph.D., Case Western Reserve University

Committee Member

David Aron, MD, MS, Case Western Reserve University

Committee Member

Yunmei Wang, Ph.D., Case Western Reserve University

Date of Defense

March 26, 2020

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Dedication

The following work is dedicated to my husband and best friend,

Timothy Edward Johnson,

for supporting me in every way on this journey.

Remembering our walks in London.

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You have all been an important part of creating this shared reality of ours.

Cross-Functional Project Team Performance: Inquiry, Identity, and Shared Reality

Abstract

By

SUSAN LEIGH JOHNSON

Firms are challenged to achieve organizational goals in an environment of increasingly decentralized information. Cross-functional project teams are employed widely as a strategy to facilitate better coordination, yet projects still fail at a rate of 31% per year. Communication is a leading cause of failure. While inter-team communication has been studied extensively, less is understood about the intra-team communication of a cross-functional project team.

The main finding of our study is that the success of a cross-functional team is dependent on the team's ability to inquire across multiple knowledge boundaries in a way that develops an awareness of each other's functional identity. Functional identity is defined as the norms and practices of a functional team which represent how they think about and prioritize their work. Whether or not this functional-identity knowledge-sharing process occurs determines whether a cross-functional team is able to construct a shared reality with respect to the projects' goals and priorities. Achieving a shared reality is what enables a team to perform successfully. We call the understanding of another's functional identity, constructed through a process of inquiry by the project team's members, their achievement of interpretive symmetry.

Our findings are from an integrated mixed-methods study. Qualitative results from Study 1 began with the consideration that cross-functional team members live in two social worlds, that of the project team and that of their own functional team. Boundaries on a project exist both from a knowledge and a social membership perspective. Therefore, team members must engage in a process of inquiry across these boundaries. We found that successful teams have a receptive awareness that the project team does not “know” and needs to learn. This receptivity supported the team members in their open inquiry with one another and the sharing of not only functional knowledge but functional identity. Exposing these elements proved necessary for the project team when negotiating a shared reality that was essential to their project’s success.

In Study 2, we sought to validate how functional identity positively influenced performance. Using a quantitative survey, we found that functional identity was mediated by the ability to learn and understand someone in a different function which increased project team performance. We noted that functional identity alone did not affect a project team’s performance. Together with findings from Study 1, we concluded that the acts of eliciting and verbalizing a functional team’s practices allow team members to share their functional identity, and this, in turn, leads to performance.

Continuing on this line of inquiry, we sought to understand through analysis of cross-functional team dialogue how functional identity played a role in the construction of shared reality. In Study 3, we found that teams with shared reality were accepting of explanations of functional identity while teams without a shared reality questioned identity. Whether functional identity was accepted or questioned was linked to the team’s shared reality or lack of one surrounding their project goal.

Shared reality is a relational theory that considers the interrelationships among the pieces within the organizational structures we create. Functional team members exist in relationship to the firm and its purpose as part of their own ongoing reality and identity. Project goals are formed to serve a firm's purpose. We found that when a project goal was not consistent with firm purpose, identity is called into question because of this ongoing interrelationship between organization and identity. A lack of alignment between goal and firm purpose prohibits the team from developing a shared reality. This was evidenced by the questioning of identity in project teams without shared reality.

In summary, as the theory of shared reality instructs us, everything exists in relationship to each other. In the case of this mixed-methods study, we find that the project team, the functional team, and the organization have an orientation to one another that is important to understand for the purpose of developing a shared reality. We demonstrated how the surfacing and communication of functional identity played a central role in uncovering this orientation and constructing a shared reality for the project team that supported their success.

In constructing a shared reality, the project team must not only take into account their relationships across their functional boundaries but must also contend with their relationship to the firm as well. Project goals that are founded in firm purpose are an additional crucial component needed to construct a shared reality. This study contributes to our knowledge on intra-team communication, cross-boundary teaming, functional identity, and shared goals, and provides empirical evidence for shared reality.

Keywords: cross-functional project teams; cross-boundary communication; shared goals; shared reality; functional identity

CHAPTER 1: INTRODUCTION

In the absence of clarity, people start to formulate their own stories. You had teams and team members with their own stories. In their heads, their own tape playing over what maybe other people might feel or whether or not they were bought in or whether or not they were going to be successful or whether or not they actually agreed. — Operations Manager and Member of a Cross-Functional Team without Shared Reality

As our world becomes more complex, information and knowledge are increasingly distributed across the firm, fragmenting functional knowledge, and as a result, firms are reliant on cross-functional teams to implement new initiatives within the organization (Anbari, Bredilett, & Turner, 2008). Given this fragmentation, the use of a project-based structure has become an essential coordinating mechanism for firms that grapple with the organizing challenges of creating and delivering new processes and products (Söderlund, 2011; Whittington, Pettigrew, Peck, Fenton, & Conyon, 1999).

Research Problem

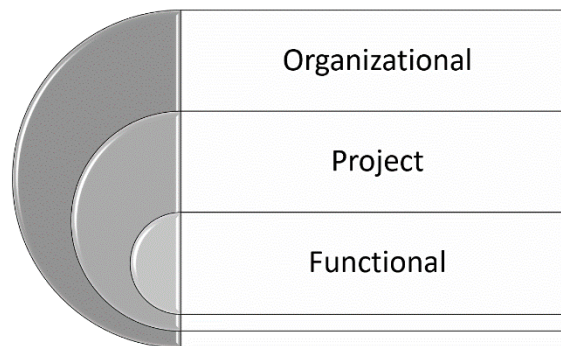
Despite the utilization and evolution of project methodologies, 31% of projects fail to meet their original goals (PMI, 2017, 2018). It has been estimated that for technology projects firms lose \$3 trillion annually as a result of this failure to meet their goals (Krigsman, 2012). In a survey of project managers, communication challenges are the most frequently cited cause of project failure (PMI, 2013). Effective communication as a firmwide capability has been shown to have a direct relationship performance. Firms with effective communication capabilities are 2.1 times more likely to outperform their peers based on their ROIs (Towers Watson, 2012).

Projects function as a central facility through which the firm's purpose and goals can be realized and through which change occurs. In the absence of effective

communication within the firm, cross-functional project teams may struggle with understanding goals and each other in their attempts to meet the objectives of the firm. A lack of understanding leads project teams to engage in a sensemaking process to establish meaning (Weick, Sutcliffe, & Obstfeld, 2005). In the absence of effective communication, sensemaking continues and can result in disjointed ideas about the goal of the project and the reality it entails (Higgins, 1992).

To understand communication in a cross-functional project team, we considered the levels of interrelationships faced by the team in achieving their goal. We approached this communication problem by formulating a view that was sensitive to the nature of the layered relationships that are inter-related in the communication process as presented in Figure 1. What follows is a discussion of a cross-functional project team's communication structure and the level at which team performance has been studied.

FIGURE 1
Levels of Interrelationships



Intra-team communication structure. We propose that the structure of a cross-functional project team more closely represents an intra-team relationship and form of communication. We reason this is the nature of the cross-functional project team

relationship because team members are divided between two worlds where their primary relationship and identity is with their functional team.

Functional teams are where individual team members share the same expertise and for which their membership is permanent. The secondary relationship of a functional team member is with the project team. In this case, members do not share the same expertise, and their membership on this team is temporary. Therefore, communication occurs between functional teams on a project through their interrelationships within the project structure across boundaries of their functional team and their project team.

The cross-functional team structure is similar to a heterarchy which has a flat hierarchy and distributed accountabilities (Kellogg, Orlikowski, & Yates, 2006). A heterarchy is characterized by collaboration across a diverse group of experts within an organization with decentralized decision making and accountability (Kellogg et al., 2006). We argue that both the relational boundaries of the functional and project team and the heterarchical structure creates distinct communication challenges when firm goals must be interpreted by the project team.

In addition to their membership and relationship to the functional team and the project team, team members have a relationship to the firm. This relationship to the firm is grounded by an individual's interpretation of the firm's purpose and how they relate to it. This relationship is, in essence, part of one's identity. Project goals serve to fulfill the objectives of the firm and, as such, support and sustain one's belief about the firm's purpose and an individual's identity. Therefore, project goals serve an important role in supporting communication across functional boundaries in a project as long as the goal is consistent with the firm's purpose.

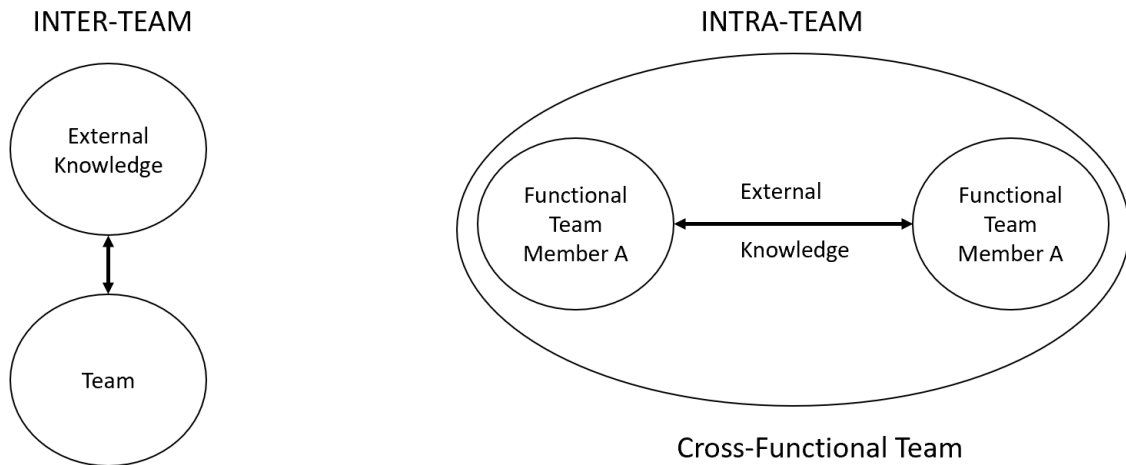
As new firm goals are established and project teams are formed to fulfill them, new routines and normative practices within the organization must be created in order to achieve organizational goals. This process of establishing new norms brings to light how the purpose of the firm, as it is represented by project goals, becomes instantiated in organizational processes and routines. The individual, the functional team, and the organization become intertwined and, as such, create expectations about the nature of their relationship to one another.

External knowledge and performance (project level). The study of teams has traditionally focused on inter-team dynamics and what enables team members to work well together. Gladstein (1984) and colleagues designed a study to conduct a comprehensive review of all the thus far known inter-team dynamic measures to consider what effect they had on a team's performance. Her study surprisingly found that not a single factor such as skills, role and goal clarity, training and rewards were directly affecting the performance outcome of the team. What her study did reveal was that teams who accessed and used external knowledge by boundary spanning performed better.

This study's finding of accessing external knowledge was within an inter-team context, and we have established that our cross-functional project team operates within an intra-team structure. Therefore, when we consider that accessing external knowledge leads to performance in an intra-team context, we mean that team members from one functional discipline are able to access knowledge from a team member in another functional discipline. While we recognize that the cross-functional team will also benefit from accessing knowledge that is fully outside the functional team members' domain—for example, from external consultants—we need to emphasize here that this same

performance benefit can be achieved within the project entity itself due to its unique structure. We present Figure 2 to more fully describe the difference between an intra- and inter-team's relationship to external knowledge.

FIGURE 2
Inter-Team vs. Intra-Team



With regard to gaining access to external knowledge, Edmondson (1999) found that in the presence of psychological safety, provided to team members by their managers, individuals would take risks to learn from others and achieve better performance. This research was conducted within a hierarchical organizational setting, while our organizational context is heterarchical. Our purpose, given this context, is to learn more about how this structure of a cross-functional project team either promotes or hinders communication across boundaries.

Identity and performance (functional level). Establishing sharedness, whether it be goals or knowledge, is particularly challenging for cross-functional teams because of the divergent knowledge worlds in which they live. Cross-functional project teams experience communication challenges because they do not share the same expertise and

the common language associated with a given discipline (Bechky, 2003). Productively integrating diverse pools of knowledge (Argote & Ingram, 2000) on a project, crossing knowledge boundaries, is a communication challenge but essential to project success. Team researchers have noted that differing expertise can create knowledge boundaries (Bechky & Chung, 2018; Edmondson & Harvey, 2018). The deeper and more specialized knowledge in a functional area, the greater the knowledge boundaries (Carlile, 2002).

Studies to determine whether differing expertise helps or hinders a team performance have proven inconclusive (Mitchell et al., 2015). While some research has found that functional diversity decreases their innovativeness (Ancona & Caldwell, 1992b), other research has found that functional diversity increases team performance (Bell, Villado, Lukasik, Belau, & Briggs, 2011; Jehn, Northcraft, & Neale, 1999). Fundamentally, we know that cross-functional teams are teams with diverse knowledge, but how an individual internalizes this knowledge difference is a matter of identity.

Taking our intra-team setting into account, we are looking at the relationship between one functional team and another as they seek to communicate. Thus, our concern is outside of the inter-team context in which functional diversity has been studied and where membership is within the same group. Our interest is in understanding how teams of teams are able to successfully navigate the knowledge boundaries of their disciplines. An individual's group identity with their functional team stands outside the project team walls.

Shared goals and performance (organizational level). The purpose of a project team is to achieve a certain goal for the benefit of the firm's mission. How teams come to an understanding of their goal is a central question. In the literature, Carson, Tesluk, and

Marrone (2007) defined shared purpose as the ability of team members to achieve a similar understanding of a goal and their engagement to achieve that goal. In their research on 59 consulting teams, they established a link between shared purpose and team performance as rated by clients.

While Carson et al. (2007) measured this as the act of verbalization of the goal and corresponding planned actions, in a later study Chow and Chan (2008) defined shared goals as the knowledge that others share your sense of what is important. In her survey of 190 managers in Hong Kong firms Chow found that shared goals motivate managers to share knowledge. Together, the findings of Chow and Carson make sense in that this sharedness with others leads to sharing knowledge and achieving success.

What the literature remains silent on is how this sharing of goals is achieved. Even though research tells us that shared goals and accessing external knowledge are the pathways to successful outcomes, it remains to better understand what promotes the creation of shared goals and facilitates the accessing of external knowledge. We have previously argued that project goals are derived from some purpose a firm has as it seeks to fulfill customer needs. That project goals have a relationship to the firm as do the individuals on the cross-functional project team. From an organizational level and perspective, we seek to understand how communication is affected by this interrelationship between project team, firm purpose, and project goal.

Gaps and Research Questions

Although inter-team communication has been studied extensively, little is understood in contrast to intra-team communication (Hoegl, Weinkauff, & Gemuenden, 2004). Edmonson has defined this as cross-boundary teaming, and in her meta-analysis

has stated that our understanding of how cooperation is achieved is limited, calling for better empirical studies and theoretical models (Edmondson & Harvey, 2018).

In the scholarly domain of coordination, researchers have noted that there has been no consideration of how one's functional identification affects one's ability to develop a shared expectation in achieving organizational goals (Bechky & Chung, 2018). We consider that the conflicting results from the diversity literature and the lack of research on shared goals support this charge. Additionally, researchers on team diversity have determined that results are inconclusive as to whether functional diversity is beneficial to or a hindrance to a team's performance (van Knippenberg, van Ginkel, & Homan, 2013).

Informed by what we currently understand about what creates team success and the diversity which may or may not hinder it, we sought to learn more about cross-functional project team communication by exploring the questions about the antecedents to the success factors of accessing external knowledge and achieving shared goals. We also sought to better understand the role of functional diversity since results have proven contradictory as to whether this may also be a source of successful team outcomes. Given these objectives, we formulated the following three research questions to advance our knowledge on cross-functional project team communication.

Research Question 1 (Project Level): What enables cross-functional project teams to share their functional knowledge with each other? The purpose of this question is to help us explain what the initial starting conditions are for this essential sharing of external knowledge that will support a cross-functional project team's successful outcome.

Research Question 2 (Function Level): Does an individual's functional diversity affect their ability to access external knowledge from another team member? The purpose of this question is to help provide additional perspectives on both functional diversity and accessing external knowledge in achieving a successful outcome in a cross-functional project team.

Research Question 3 (Organization Level): What enables a cross-functional project team to develop a shared goal? The purpose of this question is to help gain additional knowledge on how team members come to share in their understanding of the meaning of the goal in a cross-functional project team.

Research Design

The research was conducted in three phases following a mixed-methods sequential exploratory design using a QUAL → QUAN → QUAL (Creswell & Plano Clark, 2017; Teddlie & Tashakkori, 2009). An exploratory sequential mixed-methods approach was selected for the primary purpose of clarifying the factors that influence a cross-functional team's ability to communicate effectively with one another to achieve successful outcomes when seeking to achieve firm goals. Our qualitative study focused on the overall experience of cross-functional team members. Through this exploration and analysis and consultation of the literature, we found a high-level theme in constructivist theory and the expression of a shared reality within a functional team.

By comparing and contrasting successful and unsuccessful experiences within the shared reality context, we were able to draw additional related observations, again tying in the literature to understand the role of functional identity within shared reality. At the end of each phase of this mixed-methods study, we planned an integration point which

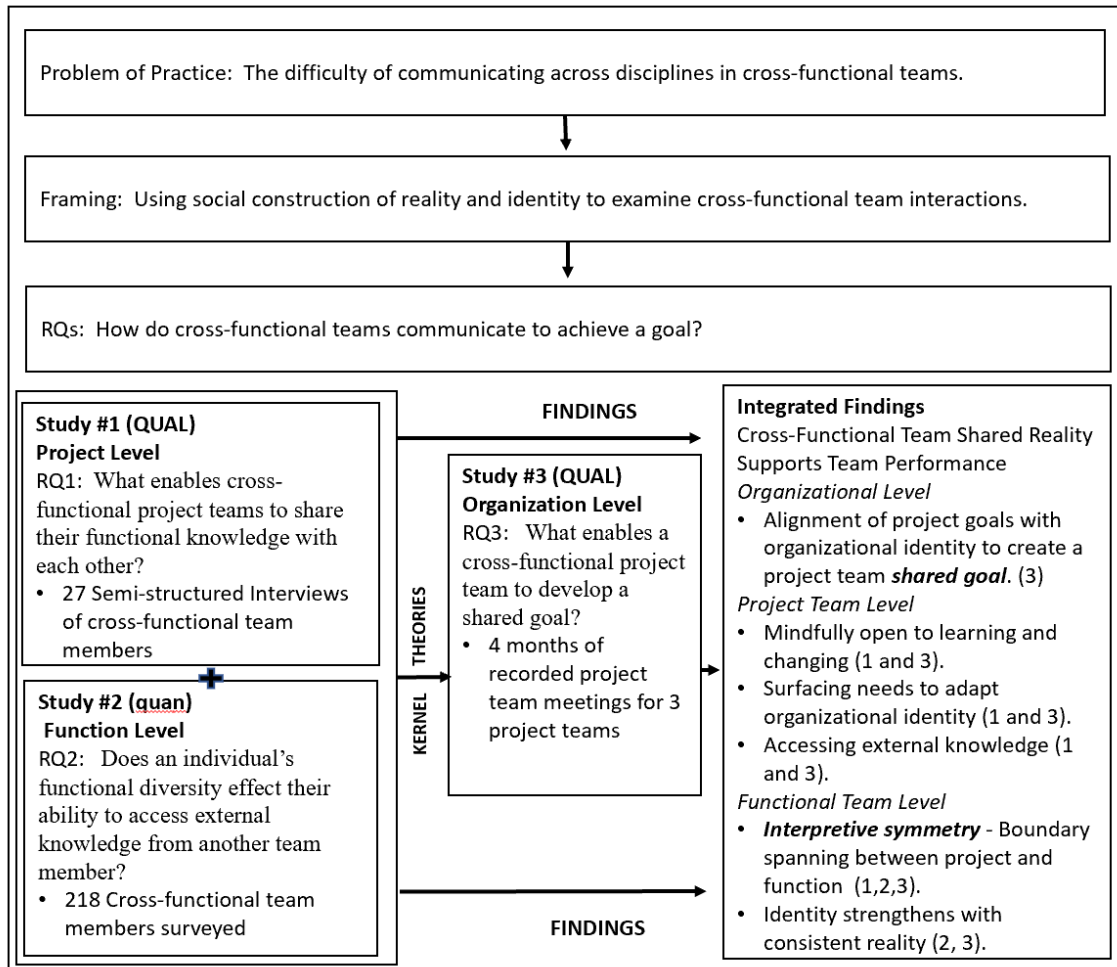
provided us with the opportunity to explore themes that emerged in the new study within the context of the previous study.

The results of the qualitative study informed the factors for the quantitative study and the results of these two studies together. In the quantitative study, we observed that a functional team's strong social identity moves in concert with a successful project outcome when mediated by learning from someone in another functional discipline. Successful project outcomes were measured as the survey respondent's perception of outsider's beliefs about the achievement of the goal.

With the first two studies, we had evolved a research agenda that focused on the social interaction of cross-functional teams and how identity as part of a group has increased the project team's likelihood of a successful outcome. In our third study, we sought to better understand how functional identity is communicated through cross-functional team dialogue. We also sought to understand more about the interplay in the relationship between functional identity and the project goal.

The study used a case study method, which allowed the researcher to directly observe team interactions and their discussions about the goal. We continued to rely on shared reality as an overarching theory to support the analysis of the cross-functional team's dialogue. The procedures for the sequential explanatory mixed-methods design of this study are presented in Figure 3.

FIGURE 3
Mixed Methods Design



Having used this mixed-methods approach, each study provides its own unique contribution, but we will also show that each contribution serves to validate the findings in the other study, which occurred at a different level. Together, our studies provide a multi-level view into how cross-functional project teams interact to communicate and facilitate the creation of a shared reality. This study demonstrated that at the functional level, a team's identity and the role they fulfill stands in an expected relationship to other functional teams and to the firm as represented by their project goal. We find that when

project goals are not aligned with the firm's purpose, a team member's interrelationship within the system is broken. This creates identity validating behaviors among the team and the propensity to project alternate realities until the goal can be aligned. We begin with a review of team literature to clarify major team concepts that are and are not related to communication in a cross-functional project team context.

CHAPTER 2: LITERATURE REVIEW

Intra-Team Communication

Because our unit of analysis for this study is the cross-functional project team, we consider first what is known about intra-team communication from this perspective. In Table 1, we present some of the key findings on intra-team communication.

TABLE 1
Intra-Team Communication

Cited Research	Summary	Theme
Gerpott, Lehmann-Willenbrock, Voelpel, and van Vugt (2019)	A study of emergent leadership on 42 self-managed project teams of university students in Germany found that change communication was important at the beginning of a project, relations-oriented communication was important at the end of a project, and task-oriented communication was always important.	Temporality and Type of Communication
Mastrogiacomo, Missonier, and Bonazzi (2014)	Developed a conversation tool based on the theory of joint activity and tested with three firms over four years. Validated that communication and conversation are coordinated activity, and a pragmatic application is to ensure conversations occur in a timely manner.	Temporality
Lovelace, Shapiro, and Weingart (2001)	A quantitative study of 43 teams from 13 high-tech firms showed that task conflict could be communicated collaboratively, leading to innovation on the product team.	Conflict Communication
Pinto and Pinto (1990)	A survey of 262 members of 72 hospital project teams found that highly cooperative teams use more informal communication while both cooperative and non-cooperative use the same level of formal communication.	Informal Communication

This view of the communication literature from an intra-team perspective considers the timing, topic, and style of communication from the perspective of the teams who engage in the communication process. This stream of literature focuses on how to enact effective communication, which, in turn, creates a positive outcome for the team. We next consider communication from the more narrowly defined context of boundary spanning where research seeks to explore the specific problem of knowledge boundaries.

Knowledge Boundaries

We understand that as a result of the project, successful cross-functional project teams will be developing new knowledge that will allow the team to achieve their goal. Participants will draw from their existing knowledge as a basis from which they seek to establish new knowledge (Blumer, 1986: 20). Thus, effective communication between functionally disparate members is central to creating this new knowledge. The ability to access knowledge that is external to a team has been shown to be a key ability of successful teams (Ancona & Caldwell, 1992b).

Current theorizing on the knowledge of teams (Nonaka, 1994) and the development of cognitive frames such as shared mental models (Mohammed, Klimoski, & Rentsch, 2000) and transactive memory systems (Lewis, 2003) has helped researchers to expand our understanding of how teams cooperate in developing shared knowledge from an inter-team perspective.

Project teams studied in these cognitive research streams are either organizationally aligned within the same functional skillset, or when diverse functional membership is required, project members are organized under a hierarchical structure. Conversely, members of cross-functional project teams in this study exist within a heterarchical project structure where reporting and accountability is matrixed. This creates unique communication challenges when there is no direct line of authority within the project. We turn then to the literature on cross-boundary teaming which has its roots in boundary spanning and is concerned both with the cognitive and identity elements of team interaction.

In the team literature, boundary spanning is defined as the team's actions to build and manage interactions with those that are external to their environment (Ancona, 1990; Ancona & Caldwell, 1992a; Malone & Crowston, 1994). Boundary spanning is, therefore, an act that requires the bridging of the discreet realities of an internal and external environment. In the process of interaction across an external boundary, there is a negotiated interpretation and adaptation to make the objects of the external environment real to the internal environment.

In design theory, Simon (1996) uses the term "appropriate representations" to broadly describe what helps individuals engage in communication when the final product is not yet known. For example, one type of representation is a boundary object which supports the coordination and integration of ideas across disciplines and facilitates the ability of multiple teams from different disciplines to align around their work (Star & Griesemer, 1989).

Star studied teams of scientists and identified the concept of boundary objects as the means by which translation can occur across social worlds (Star & Griesemer, 1989). In her study, a boundary object, such as a map, allowed two researchers from different disciplines to know what something was, but at the same time allowed them to interpret it differently. This allowed the team to communicate, collaborate, and develop a common finding. The creation and utilization of boundary objects represents a key capability in developing and maintaining coherence across teams with functionally diverse expertise (Bowker & Star, 1999).

Thought has been considered as a boundary object which is subject to differing interpretations yet shareable in the process of perspective making and taking (Boland Jr

& Tenkasi, 1995). Here, knowledge or thought rather than physical objects which can be known by individuals support boundary spanning when functional backgrounds differ (Boland & Tenkasi, 1995).

Cross-boundary teaming is defined as a team's actions to establish linkages and manage interactions with parties in the external environment (Ancona, 1990).

Edmondson and Harvey (2018) further refined the definition by specifying that this is not the domain of cross-functional expertise embedded in a stable, self-contained team but related to teams that are temporary. This stream of research addresses the intra-team communication structure we are studying. We present literature in Table 2 as some of the key research on what is known thus far about boundary spanning and its relationship to cross-functional team performance.

TABLE 2
Communication and Knowledge Boundaries

Cited Research	Summary
Bechky and Chung (2018)	An ethnographic study of an equipment manufacturer and four film production projects found that occupational control processes/normative practices supported coordination, but the expectation of others about the norms differed. The film crew members expected quality while the manufacturing members expected reputational protection.
Seidel and O'Mahony (2014)	A case study of six projects and their ability to create novel innovations found that the use of a boundary object and the act of collectively reviewing and editing it given design constraints lead to concept unity in the form of concept coherence.
O'Mahony and Bechky (2008)	A quantitative study of 31 Consulting Teams of five to seven MBA students. Increasing boundary spanning reduced role overload and improved individual and team effectiveness.
Kellogg et al. (2006)	An ethnographic study of 110 employees of an advertising firm that worked on project teams. Finding that in order to share knowledge, individuals follow a process of display, represent, and assemble. Uses the concept of trading zone to describe how meaning is negotiated across knowledge boundaries.
Carlile (2002)	A one-year ethnographic study of teams in an automotive parts manufacturing facility found that knowledge is both a barrier and a source of innovation.
Ancona (1990)	A qualitative study of cross-functional project teams engaged in restructuring an educational institution. Finding that external boundary spanning leads to increased team performance.

Studies on boundary spanning have a long history while the workplace dynamics continually evolve to increasingly rely on this core capability of a team. Research into the use of boundary objects or boundary spanning practices provides us with an evolving view of how organizations in different settings manage to coordinate their diverse knowledge in order to achieve positive outcomes. The need to continue to build our understanding of how knowledge is shared is recognized by scholars in many streams of literature that touch on this issue from diversity to coordination to boundary spanning, much still needs to be learned.

While knowledge boundaries have been considered from an actions perspective thus far, there is also a significant literature stream that is addressing boundaries from the individual actor's perspective and how their sense of group belongingness can promote or inhibit coordination across this boundary. We next address the literature on social identity as it relates to the disciplinary specialization or functional expertise in a cross-functional project team.

Team Identity

Perceptions of group membership in a project context is an important factor as it has the ability to influence how functionally disparate individuals interact. Projects have been defined along many different dimensions, but we now must address the social context of a project (Anbari et al., 2008). Diversity researchers have defined functional experts as individuals with differing informational resources, experience, and social network ties (Argote & Ingram, 2000; Jehn et al., 1999; Mitchell et al., 2015). When a functional expert joins a project, they become a member of a cross-functional team (Denison, Hart, & Kahn, 1996; McDonough III, 2000). Cross-functional team members

on projects are challenged to span two social worlds, that of their project and that of their function.

Ellemers, Kortekaas, and Ouwerkerk (1999) found that the ability of an individual to actively engage in the selection process of group membership will influence their affective commitment to that group. Functional team members have typically chosen their career and the company and department in which they work. For a project participant, this level of agency does not necessarily exist because individuals are customarily assigned to work on it by their manager. Therefore, this lack of agency in being able to choose to participate in a project reduces an individual's sense of identity with that project.

In this same study, Ellemers found that group size affects our cognitive sense of membership. Projects that cross boundaries are not only engaging an assigned individual from a specific functional team; they are, in essence, seeking to engage the knowledge of the entire functional team by surfacing knowledge from whomever it resides within. Therefore, the functional participants on a project will, in actuality, be large and diverse through this extension from the assigned functional team member to their connection with the wider functional knowledge of their team. We argue that the group size and continuously shifting participants further weakens an individual's sense of identity or "teamness" with other individuals who participate in a project.

In a meta-analysis by Riketta and Van Dick (2005) on team versus organizational identity, the authors found that individuals are much more strongly tied to their workgroups than they are to their organizations. Given the findings of Ellemers on individual agency in team selection and the effects of team size along with the relative

strength of an individual's workgroup identity and the temporary nature of project work, we argue that an individual's membership in their functional team will be a more salient identity than their identity with project members.

Given that we have established that functional identity will be most salient for a project team member in this capacity, they are therefore compelled to act as a boundary spanner. Boundary spanners operate at the interface between groups that share identity as well as specific knowledge or capabilities (Schotter, Mudambi, Doz, & Gaur, 2017). When we are concerned with crossing a boundary of understanding the nature of the social relationship is essential to the concept of the cross-boundary entity.

In the case of a cross-functional team member, we have the identity of the project team and the identity of the functional team. To our knowledge, there have been no studies that consider both identities in the cross-functional team context, and the perspective is either from a superordinate identity of the project team or a functional identity perspective. The underlying supposition in one case is that the superordinate identity of the project team subsumes the functional identity or "diversity" of the team members. Alternately, researchers consider identity from the functional perspective and consider boundary spanning between identities and how one gets past the deficit of a functionally identified diverse team. We present research that relates to identity from these two perspectives at an intra-team level in Table 3.

TABLE 3
Knowledge Boundaries and Identity

Cited Research	Summary	Identity Orientation
Mitchell et al. (2015)	A quantitative study of 75 teams in an Australian hospital found that leader inclusiveness increases shared team identity and a reduction in perceived differences increasing performance.	Superordinate
Kane (2010)	A quantitative analysis of 216 students simulating a production line to make origami paper boats found that superordinate identity supported knowledge transfer.	Superordinate
Richter, West, Van Dick, and Dawson (2006)	A quantitative study of 53 workgroups in five healthcare organizations. For boundary spanning individuals, organizational identity moderates the relationship between workgroup identification and effective intergroup relations. If workgroup identification is high and boundary spanning is high, then workgroup identity was most strongly related to effective intergroup relations.	Functional
Van Der Vegt and Bunderson (2005)	A quantitative study of 57 engineering teams working in a global oil and gas company. Measured performance based on supervisor rating. Measured collective affective team identity and found that this moderated the relationship between functional diversity and performance.	Superordinate
Sethi (2000)	A quantitative survey of 118 project managers of cross-functional project teams in consumer products companies. Superordinate identity positively influences the performance of new product development teams. Outcome interdependence and autonomy increase superordinate identity. Functional identity reduces autonomy and the effect on superordinate identity.	Superordinate

In summary, the perspective identity research has been on either how a superordinate identity is established as a beneficial strategy for achieving a positive project team outcome or how functional identity can be mitigated in order to support a positive outcome.

Shared Goals

There has been relatively little research devoted to the understanding of how teams conceptualize or construct and test the existence of a shared goal. We present the extant research on shared goals in Table 4.

TABLE 4
Cross-Functional Teams and Goals

Cited Research	Summary
Chow and Chan (2008)	A survey of 190 managers from Hong Kong firms found that shared goals contributed to a person's determination to share knowledge. Shared goals are defined as the extent to which the team has shared objectives.
Carson et al. (2007)	A survey of 59 consulting teams found that shared purpose leads to team performance as measured by the client. Shared purpose is defined as discussing goals, associated tasks, and making plans to achieve the goals.
Knight, Durham, and Locke (2001)	Using a computer simulation with teams found that team performance was positively affected by goal difficulty.
McDonough III (2000)	A meta-analysis on cross-functional team literature found the most often cited factors for effects on team performance were: (1) appropriate goals, (2) decision making authority, (3) appropriate resourcing, and (4) a productive climate
Pinto, Pinto, and Prescott (1993)	A survey-based quantitative study of 131 hospital new product development teams found that the presence of superordinate goals led to successful outcomes as perceptions about achieving both a task outcome and a psychosocial outcome of a positive team environment.

In research on social capital, a shared goal was defined as “My organizational members and I always agree on what is important at work” (Chow & Chan, 2008). While this definition captures a sense of a shared prioritization, the question does not capture the goal itself or how the team understands the goal.

In an earlier quantitative study of project team members from 131 hospitals developing new programs and services, Pinto found that superordinate goals lead to enhanced cooperation (Pinto et al., 1993). While a full scale was not provided for superordinate goals, an example question is whether all team members are committed to the goal. While this captures goal commitment, we cannot know if other measures would capture whether the team had a shared belief about the goal.

Alternately, on the topic of leadership, the construct of shared purpose is defined as the team members having a similar understanding of their objective in pursuit of their goals (Carson et al., 2007). In their development of a measure for this construct, Carson

et al. (2007) defined shared purpose as the acts of discussing goals, ensuring shared understanding and devising action plans to achieve these goals.

We find this measure consistent with the definition of a goal as an end towards which effort is directed (Merriam-Webster, n.d.). This study of 59 consulting teams found that shared purpose leads to successful outcomes as evaluated by the client establishing the relationship between shared goals and success. While this study defined shared goals and their effects, it did little to elucidate what factors affect whether a team is able to achieve that shared understanding of the goal.

We argue that a goal as situated in an organizational context and as an object of reality contains not only the end which is being strived for with that goal or the “what,” but it must also be motivated by reasons of “why,” which is essential to the goals meaning. Here, note that when crossing boundaries, we assume the object of the goal is standing independently as an object with its own reality. Hence, in crossing knowledge boundaries, we are proposing a new way in which we can think about how a goal is shared and how individuals enact their relationships with it.

CHAPTER 3: THEORETICAL FRAMEWORK

Shared Reality

Bruner's (1957) discussion on the process of perception stated that we observe and then categorize the world we experience. In this process, we examine the features of an object and then seek to categorize the object so that we can ascribe meaning to it based on the qualities or distinctions associated with its classification.

Berger and Luckman (1991) defined these objects using the term "reality," which is a phenomenon that has a being that is separate from our own (p. 1). They distinguished the reality of objects from knowledge, defining knowledge as the acceptance that the phenomena is real and has certain associated characteristics (Berger & Luckmann, 1991: 1).

Echterhoff, Higgins, and Levine (2009) defined the term reality from the perspective of psychology as one's subjective experience that something is true, and they note that this truth exists for an individual regardless of its verifiability. We, therefore, use these ideas of a specific object or phenomena which is perceived as truth as the meaning of reality.

Communicating meaning is not achieved directly but is a social process, and meaning is negotiated through one's sharing in the construction of reality with other humans (Berger & Kellner, 1964; Berger & Luckmann, 1991; Bruner, 1991; Echterhoff et al., 2009). The beliefs and truths of our reality are importantly connected with other members of the group (Bar-Tal, 2000). The definition of "shared" in the term shared reality is beliefs or feelings that are held in common among a specific group (Echterhoff

et al., 2009). Whether or not a group is able to form a shared reality is dependent on the process of communication they employ.

When we engage in interaction with another, we enter the conversation with preconceived patterns of how we characterize that individual and their knowledge. In face-to-face interactions, these preconceived patterns cannot hold as the individuals continuously express themselves in ways that contradict our patterns (Berger & Luckmann, 1991: 30). This process of continually adapting one's perception of reality has lead organizational psychologists focused on team research to converge on a conceptual view of teams as complex, adaptive, dynamic systems (McGrath, Arrow, & Berdahl, 2000).

Symbolic systems. Vygotsky's (1962) key insight on social interaction is that we use both language and symbolic systems to shape the reality we present to one another. While reality is shaped by the systems of communication we rely on, it is also shaped by the individuals with whom we must communicate. The extent to which we shape reality by considering the person we are engaged in a dialogue with determines how successfully we can achieve a common interpretation (Higgins, 1992). The consideration of another's view includes what the other's expectations may be regarding how individuals in the group perform their roles (Higgins, 1992). The consideration of another person is often referred to as stepping into another's shoes. Importantly, Mead notes that through this enactment of considering another, we can establish that there is a self that is separate and distinct (Blumer, 1986).

Wittgenstein (1974) explored the unique, complex, and varied ways in which we communicate meaning in his analysis of our use of language. His philosophy of language

provides us with insight into the challenge of communication, particularly as it relates to knowledge boundaries. In his ideas about language games, Wittgenstein proposes that the words we use carry with them their own narrative history and purpose that is uniquely situated within a context (Astley & Zammuto, 1992).

In the domain of organizations, history is shared across the membership of a functional team and is below the surface of consciousness. This history and context associated with words does not extend beyond the team's boundary. How to navigate this absence of context or surfacing that context adds to the challenge of communicating across boundaries. We, therefore, must consider ways in which contextual knowledge may be surfaced.

Through the act of speaking, things become more real to us. This is because we are able to think and speak at the same time in an intersubjective exchange, and this verbalization exposes our reality more solidly to us and to the other (Berger & Luckmann, 1991: 38). Language provides us with the ability to span discreet realities (Berger & Luckmann, 1991: 39)

Bruner (1991) considered the verbalized construction of reality from the perspective of narrative, which serves as a means to organize our experience. An essential part of our stories created through narrative is the story's ability to provide justification or reasoning for our perceptions and our acts. In a cross-functional team setting, the stories and realities originating from different functional worlds must be rationalized and merged with the story that is evolving within the project team. The process of individuals rationalizing functional knowledge within a project context is akin to the sensemaking process in high-reliability organizations but more complex when

there is not an existing shared process as a referent as exists in an HRO (Weick et al., 2005).

Uncertainty and interpreting. In the process of our daily existence, we continuously encounter unexpected events that, in some way, affect our negotiated reality and may give reason for us to examine it. We refer to the extensive literature on routines that acknowledge we have an innate sense of urgency to make sense of our world and rectify inconsistencies (Dionysiou & Tsoukas, 2013; Feldman & Rafaeli, 2002; Weick et al., 2005). From the perspective of a shared reality, these events expose not only our objects of reality but our identity in relationship to those objects.

The interpretation of events is not limited to the particular facts and circumstances of the event in isolation. When we are subjected to environments and experiences that are prone to uncertainty, developing a shared reality supports one's ability to build certainty through this process of social verification (shared reality) and corroboration with others (Levine, Higgins, & Choi, 2000). Baron (1996) argues that the corroborative interaction is so frequent and ubiquitous that of necessity, it does not require that we know or consider each other's values and norms, nor does it require that this information be integrated into our own.

By considering ourselves and the other in our representation and negotiation of reality, we are, in essence, binding that individual or group of individuals and ourselves in relationship to that object of reality and to each other. This interlinking of the individual and the object in our experience creates an opacity such that "Identity remains unintelligible unless it is situated in a world" (Berger & Luckmann, 1991: 174). Therefore, it is through our objects of reality that we find not only meaning for the object

but through which we can come to know our own self in relationship to both the object and to others.

Social Identity

Tajfel (1984) defined perceptions of group membership as social identity in which “that part of an individual’s self-concept” that is derived “from their knowledge of their membership of a social group” along with “the emotional significance” they attach.

Ellemers extended this definition to encompass three measurable components associated with a group identity; affective (feelings about group membership), cognitive (knowledge of membership in a group), and evaluative (the status of the group) (Ellemers, Van Rijswijk, Roefs, & Simons, 1997).

Identity occurs at multiple levels; there is intra-personal at the individual level, interpersonal at the dyad level, intragroup at the team level, and intergroup at the between team level (Ellemers, 2019). Our discussion on identity is focused on the intergroup level, which is the intra-team level, and it is important to note here the key identities at play within the cross-functional project setting.

Individuals can belong to multiple groups simultaneously, and the social identity that is invoked by an individual is driven by what they sense is appropriate for the context (Kreiner, Hollensbe, & Sheep, 2006). In the project context, we see that individual participants have three group memberships that are particularly relevant. Project membership is one potentially salient identity on a cross-functional project team. Individuals on the project team are also functional experts and members of a team within a given discipline, which is another source of identity. The firm also plays a significant role in a project context as cross-functional teams seek to implement goals that are

derived from a firm's purpose. Team members have a relationship or identify with the ongoing firm purpose, which is also represented by the project goal.

Identity is not static but is continually evolving in response to the reality we continuously negotiate with others (Brown & Lewis, 2011). Moments of unknowingness can surface questions of identity such that as the environment is changing, individuals must redefine who they are in relation to those changes (Nag, Corley, & Gioia, 2007). This need to validate identity is a relational exercise in which we engage with the relevant sources of definition in our environment to clarify our own sense of self. This exercise is not limited to our own self but extends to how we understand or change our understanding of others in our environment. This process of continuous redefinition has been defined as "identity work" (Alvesson, Lee Ashcraft, & Thomas, 2008).

A large body of research on group identity has focused on the nature and causes of intergroup conflict. The term intergroup threat is used and is defined as when the goals of one group are threatened by the beliefs, actions, or characteristics of another group (Riek, Mania, & Gaertner, 2006). It is important to note that in this line of inquiry, it was found that when goals are complementary, then intergroup relations improve and when goals are not aligned, they deteriorate (Sherif & Sherif, 1969).

Social Construction

The theoretical concepts of shared reality and social identity are brought together through the ongoing process of social construction. Social construction refers to the process of how individuals engage to negotiate their beliefs about the objects of reality and about each other. Social construction is dynamic and ongoing, and neither reality nor identity remains constant in the face of an ever-changing environment.

The process of social construction is motivated by some abstract idea of a goal for which participants must establish a shared set of beliefs. In cross-functional teams, we are considering social construction at the intra-team level. The aim of a team engaged in the construction process is to first make that idea of the goal knowable and shared by the team.

The interpretation of the goal will rely on foundational items of how an individual and a team understand their relationship to the firm. This foundational layer of identity consists of the norms, practices, and priorities the team has established within the context of the firm that have been motivated by their need to perform as functional experts. Moving the goal from the abstract into reality requires that it be shaped through and that it refers back to these items of identity.

If we use the example of building a house, we know that what the plumber thinks and cares about versus the electrician or the drywaller are different. However, the creation of the house will have embedded within it their value systems and perspectives about their individual trades in the way that it is constructed. The team of electricians may prefer a certain brand of wire because they believe it is worth the extra expense. The team may wish to work alone on the house in the absence of other trades so that they can be more thoughtful in the execution of their work. These items of preferences and values will, in turn, influence the way in which the electrical work will be executed in the house, and those values will be sustained across time as part of the reality of the house as long as the house exists.

In the more complex cross-functional team construct, we have a team of teams who are, in turn, each building their ideas about the goal and creating an object of reality.

We can now see that this cross-functional team's goal as an object will have many foundational layers of interpretation. These layers are immersed with the identities of the respective unique expertise of each team engaged in the project.

Teams must interact with each other and elicit these definitions of each team's construction of the goal. Learning the why of a team's meaning is a process of discovery needed to reveal the important interrelationships between one team's conceptualizations of the goal and another's. While we are relaying this as a step-like process, this process is actually occurring dynamically such that each team is revealing and potentially modifying and adapting how they have modeled the idea of the goal so that it can accommodate the perspective of another team. This process creates interrelationships across teams that they have uniquely constructed in relationship to the goal.

CHAPTER 4: STUDY 1 – QUAL – RECEPTIVITY AND INTERPRETIVE SYMMETRY ON PROJECT TEAMS

Abstract

This paper is based on the findings of a qualitative study of 27 cross-functional team members in financial services firms. The findings resulted in the development of a process model of cross-functional project team communication that supports the creation of new knowledge. The essential function of the communication model is that reality is constructed across functional divides in a way that results in alignment both inside the project team and outside within the organization's functionally dependent teams. We have termed this alignment across functional boundaries as interpretive symmetry. This process model builds on past research on boundary spanning. We identify boundary space as the place and the opportunity for communication to occur. We also identify mindfulness among the project team as the team's awareness and acceptance that they do not know, which opens them up to develop new knowledge. This model of communication represents this paper's main contribution to the literature on cross-functional teams. This paper also contributes to our understanding of the role of boundary spaces and mindful states in promoting cross-boundary communication.

Keywords: cross-functional teams; communication; shared reality; knowledge

Introduction

As our world becomes more complex, information and knowledge are increasingly distributed across the firm, and organizations are increasingly reliant on project teams to implement new initiatives within the firm (Anbari et al., 2008). With the growth in the use of project teams, formalized methodologies have emerged for

managing projects to increase the probability of success. However, while project success rates have improved over time, 31% of projects still failed to meet their original goals, and \$97 million in project investment continues to be lost each year (PMI, 2017, 2018).

Communication challenges are commonly cited by project managers as a significant contributing factor to project failure (PMI, 2013). Cross-functional teams have additional communication challenges because they do not share the same expertise and the common language associated with a given discipline (Bechky, 2003). Deep and specialized knowledge in a functional area can create knowledge boundaries (Carlile, 2002). Researchers on team diversity have concluded that results are inconclusive as to whether functional diversity is beneficial to or a hindrance to a team's performance (van Knippenberg et al., 2013)

A central purpose of a project is to develop new knowledge that will allow the team to achieve its goal. Participants will draw from their existing knowledge as a basis from which they seek to establish new knowledge (Blumer, 1986: 20). Thus, effective communication between functionally disparate members is central to creating knowledge. Furthermore, the creation of new knowledge has been linked to the success of teams (Ancona & Caldwell, 1992b). Effective communication among team members as they interact is, therefore, central to the team's ability to create new knowledge and achieve success. Therefore, our research question was to ask: "How do cross-functional teams communicate across functional boundaries to create shared knowledge?"

We approached the research question by conducting a qualitative study of cross-functional project team members in financial services firms. In this study, we sought out interviewees with differing functional expertise. Examples of functional expertise ranged

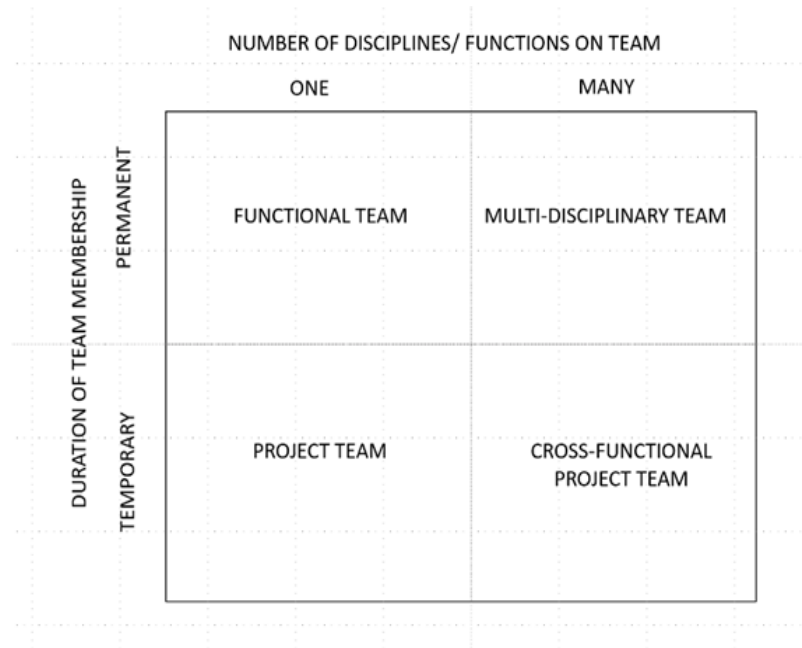
from direct operational business functions such as customer service and back-office management to support roles such as project management and technology development. The analysis focused on how interviewees experienced communication as it occurred across disciplinary boundaries.

It is important to first define our meaning of a cross-functional team. Team studies represent a large body of literature, and there has been a lack of consistency on how they have been defined (Hollenbeck, Beersma, & Schouten, 2012). Kozlowski defined the characteristics of teams as being: composed of two or more individuals, existing to perform organizationally relevant tasks, sharing one or more common goals, interacting socially, exhibiting task interdependencies, maintaining and managing boundaries, and being embedded in an organizational context that sets boundaries, constrains the team, and influences exchanges with other units in the broader entity (Kozlowski & Bell, 2003). We adopt this definition of a team.

The definition of functional and cross-functional teams, as applied in this study, is informed by the dimensional scaling framework approach developed by Hollenbeck (Hollenbeck et al., 2012). These dimensions include temporal stability and skill differentiation.

Functional teams are defined here as teams who have specific skills and experience that are not easily interchangeable with that of other teams within the organization, and membership is permanent. Cross-functional project teams are, in turn, defined as teams with representatives from more than one functional team and are formed on a temporary basis to accomplish a task. The dimensions of the cross-functional team, as used in this study, are presented in Figure 4.

FIGURE 4
Cross-Functional Project Team Dimensions



This cross-functional team structure is similar to a heterarchy which has a flat hierarchy and distributed accountabilities (Kellogg et al., 2006).

Current theorizing on the knowledge of teams (Nonaka, 1994) and the development of cognitive frames such as shared mental models (Mohammed et al., 2000) and transactive memory systems (Lewis, 2003) has helped researchers to expand our understanding of how teams cooperate to develop shared knowledge. Teams studied in these research streams are either aligned within the same functional skillset, or when diverse functional membership is required, project members are organized under a hierarchical structure. Conversely, members of the cross-functional project teams in this study are hierarchically aligned to functional teams. This creates unique communication challenges when there is no direct line of authority within the project. In the literature review that follows, we review theory on how knowledge is constructed and shared.

Literature Review

Socially constructed reality. Bruner's (1957) discussion on the process of perception stated that we observe and then categorize the world we experience. In this process, we examine the features of an object and then seek to categorize the object so that we can ascribe meaning to it based on the qualities or distinctions associated with its classification. Berger and Luckman (1991) defined these objects under the term "reality," which is a phenomenon that has a being that is separate from our own (p. 1). They distinguished the reality of objects from knowledge, defining knowledge as the acceptance that the phenomena is real and has certain associated characteristics (Berger & Luckmann, 1991: 1).

The process of creating a reality and knowledge that is shared is dependent on social interaction. When we engage in interaction with another, we enter the conversation with preconceived patterns of how we characterize that individual and their knowledge. In face-to-face interactions, these preconceived patterns cannot hold as the individuals continuously express themselves in ways that contradict our patterns (Berger & Luckmann, 1991: 30). This process of continually adapting one's perception of reality has lead organizational psychologists focused on team research to converge on a conceptual view of teams as complex, adaptive, dynamic systems (McGrath et al., 2000).

This adaptive process that forms a shared reality cannot occur without actively engaging in communication with another. Through the act of speaking, things become more real to us. This is because we are able to think and speak at the same time in an intersubjective exchange, and this verbalization exposes our reality more solidly to us and to the other (Berger & Luckmann, 1991: 38).

Language provides us with the ability to span discreet realities (Berger & Luckmann, 1991: 39). In the team literature, boundary spanning is defined as the team's actions to build and manage interactions with those that are external to their environment (Ancona, 1990; Ancona & Caldwell, 1992a; Malone & Crowston, 1994). Boundary spanning is, therefore, an act that requires the bridging of the discreet realities of an internal and external environment. In the process of interaction across an external boundary, there is a negotiated interpretation and adaptation to make the objects of the external environment real to the internal environment.

Supplementing the use of language for constructing a shared reality is the use of boundary objects. Star studied teams of scientists and identified the concept of boundary objects as the means by which translation can occur across social worlds (Star & Griesemer, 1989). In her study, a boundary object, such as a map, allowed two researchers from different disciplines to know what something was, but at the same time allowed them to interpret it differently. This allowed the team to communicate, collaborate, and develop a common finding. The creation and utilization of boundary objects represents a key capability in developing and maintaining coherence across teams with functionally diverse expertise (Bowker & Star, 1999).

Thought has been considered as a boundary object which is subject to differing interpretations yet shareable in the process of perspective making and taking (Boland Jr & Tenkasi, 1995). Here, knowledge or thought rather than physical objects which can be known by individuals support boundary spanning when functional backgrounds differ (Boland & Tenkasi, 1995).

Bruner (1991) considered the verbalized construction of reality from the perspective of narrative, which serves as a means to organize our experience. An essential part of our stories created through narrative is the story's ability to provide justification or reasoning for our perceptions and our acts. In the model of cross-functional team communication, members from different functional stories and realities must find ways to rationalize and merge their stories in order to organize their shared experience in a process we liken to sensemaking (Weick et al., 2005).

Research Design

Methodology. This research was conducted following the approach of qualitative, grounded theory, as formulated by Glaser and Strauss (2012). Grounded theory is defined as an integrative methodology. The approach is based on new theory being constructed from the study of interactions with and between people and considers these interactions from both historical and current practices perspective (Charmaz, 2014). This research method helps the researcher uncover and understand the meaning of lived experiences of the participants and apply an analytical process to derive theoretical understanding and uncover patterns and conceptual frameworks that are aimed at adding to the body of knowledge of both the academic and practitioner communities.

The research question posed to the interviewees in this study was: "My study involves members of project teams and I am interested in learning about how teams interact together in projects that are successful and unsuccessful. I would also like to understand how team's problem solve together to create new knowledge."

Sample. Interviewees were team members of cross-functional project teams consisting of project managers, business analysts, and subject matter experts with a

sample size of twenty-seven. Interviewees were intentionally selected to represent different areas of functional expertise in the sample. The purpose of a diverse sample was to represent the differing perspectives of team members in their orientation to their functional knowledge in a project setting. Those interviewed were fairly evenly distributed between male and female for each role type with the exception of development managers who were all male (Table 5).

TABLE 5
Summary of Interviewees

Team Member Role	Female	Male	Total
Business Analyst	3	2	5
Development Manager	0	4	4
Project Manager	3	4	7
Subject Matter Expert *	7	4	11
Grand Total	13	14	27

* Subject matter experts were those in the organization who performed operational roles within the business such as customer service, business managers, business operations functions and finance.

The population sampled was divided between those participants for whom project work is an added responsibility to their daily operational work, with 55% of those interviewed having a primarily operational role while 45% had primarily a project role.

As analysis progressed, the focus began to center on the successful projects and delineated them between those that had achieved success without a prior failure and those that effectively had gone through a reset caused by an initial failure(s) in order to achieve success. Details of the projects and the functions of the interviewees are presented in Tables 6 and 7. While the interview process was focused on individual experiences, our study did capture three projects where multiple interviewees discussed the same project. This provided us with the opportunity to triangulate their experiences from multiple viewpoints.

TABLE 6
Summary of Projects with Initial Success

Projects That Were Initially Successful			
No	Interviewee Role *	Company Type	Project Description
1A	OM, PM, TM	Mid-size Fintech	Build a new product that supports payment acceptance in an invoice workflow software as a service (SAS) product.
1B	BA	Large Domestic Bank	Build an accounts receivable application.
1C	BA	Large Insurer	Build robotics into an insurance claims process.
1D	OM	Large International Bank	Centralize a foreign exchange operations function.
1E	OM	Mid-Size Securities	Create an automated securities pricing service for trading clients.
1F	BM	Mid-Size Securities	Implemented a securities portfolio analysis tool for the sales team.
1G	EC	Large Mutual Fund	Development of a black box technology staffing model to accelerate technology delivery.
1H	OM	Large Bank	Implement a customer workflow tool for an existing Know Your Counterparty process.
II	OM	Regional Bank	Merger of two regional banks.
1J	PM	Mid-Size Fintech	Build-out technology infrastructure in a new building.
1K	TM	Large Domestic Bank	Merger of two large banks.
1L	FM	Mid-Size Fintech	Select and implement a new accounts receivable application.

* Roles: BA – Business Analyst, EC – External Consultant, PM – Business Manager, OM – Operations Manager, PM- Project Manager, TM – Technology Manager, FM – Finance Manager

TABLE 7
Summary of Projects with Initial Failure

Projects Where There was Failure before Success			
No		Company Type	Project Description
2A	OM, CSM, CS, PO	Mid-Size Fintech	Move clients unto a new invoice autopayment service on an application that was acquired through a merger.
2B	TM, PM	Mid-Size Securities	Project to provide internal sales desk with tool to match client's security inventory.
2C	PM	Mid-Size Fintech	Centralize and commonize data across multiple platforms.
2D	PM	Large Bank	Define Know Your Counterparty regulatory requirements into a business process and implement a system to support it.
2E	CSM	Regional Bank	Merger of two regional banks.
2F	BA	Mid-Size Securities	Enhance a compliance monitoring tool with reporting.
2G	TM	Mid-Size Securities	Update a trading system.
2H	BA	Mid-Size Securities	New product implementation requiring the build of a new trading application.
2I	(2) PM	Mid-Size Fintech	White label and customize a SAS payables application.
2J	PM	Mid-Size Securities	New product implementation requiring the build of a new trading application.

* Roles: BA – Business Analyst, CSM – Customer Service Manager, CS – Customer Service PM – Business Manager, OM – Operations Manager, PM- Project Manager, PO – Product Owner, TM – Technology Manager

Data collection. Interviewing started in May 2017 and was completed at the end of August 2017. Interview candidates were primarily interviewed face-to-face with two interviews conducted by phone. Interviewees were drawn from the researcher's professional network. Interviews were typically between 40 and 60 minutes. The interview approach was conducted in a conversational format, allowing participants to share their lived experiences related to the research topic.

Data analysis. In keeping with the principles of a grounded theory approach, the analysis of data began following the first interview and continued throughout the data collection period. The coding process was dynamic and interactive throughout the research project. Coding began with initial or open coding which split the data into individually coded segments followed by a second cycle of selective coding which was aimed at reorganizing and focusing the data. Axial coding followed, which is the process of mapping categories to subcategories and specifying the properties and dimensions of a category (Charmaz, 2014). The results of the open coding process are presented in column 1 of Appendix A.

Phase II of the research process was guided by exploring the most frequently referenced subcategories in context with the entire population. The major themes within the categories of the study are presented in column 2 of Appendix A. The cross-referencing of what had now become the major themes with existing theory occurred dynamically throughout the entire analysis. The final theoretical mapping is presented in column 3 of Appendix A.

Findings

Receptive states of the project team. Cross-functional teams who achieved a successful outcome exhibited a heightened receptivity to their environment because they were very aware and attuned to the fact that they lacked the knowledge needed to achieve their goals. Teams either began their project in a state of awareness, or they became aware through a failed prior attempt. Examples of narratives from a cross-functional team in an initial attuned state at the beginning of a project are presented in Table 8.

TABLE 8
Teams Who Are Receptive at Project Start

Receptive State – Knowing that You Don’t Know – Initially Successful Projects		
Project	Interviewee	Quote
1A	Project Manager	“It was something we had never done before, entering into the payments business. A lot of people, none of us really knew what that meant, and we were kind of figuring it out as we were going along.” “Going into a project not really knowing what you’re implementing, or probably the deep level of knowledge that you should have for what you’re implementing. Not having a choice, just you got to get it done, and you got to just try and go out there and figure it out.”
1B	Business Analyst	“There was a sense of, we are going to learn our way into the right solution.” “I thought, and I understood what the key components of this experience were... I realize that I only really understood a small part of it.” “But the really interesting piece is, here’s what you thought, and here’s how it really is. Now, how does that change the product strategy?”
1E	Operations Manager	“When I took over client pricing, I was trained from the previous person for two days. So, to understand the whole process took a while but, I was able to build it out or help build it out with the technology, based on my frustrations.”

For cross-functional teams who were in a receptive state at the beginning of the project, the awareness of not knowing seemed almost serendipitous and was established in differing ways. In the case of project 1A, the resident expert on payment processing in the firm had left the company. In the case of project 1B, a senior leader required the product team to engage in an ethnographic study with potential customers, which led the

team to an awareness that they had made incorrect assumptions. In the case of project 1C, the operations manager who had taken over a securities pricing process is new to the organization and must learn the process in order to be able to automate it. As the central subject matter expert, she must surface from the organization how things work while she is also engaging with the cross-functional team.

When team members are attuned to their environment, it opens the team up to consider new ways to approach their goal. In Table 9, we present an example interview statement that demonstrates this change of mind within the team.

TABLE 9
Teams that Become Receptive Through Failure

Receptive State – Acceptance that You Don’t Know – Initially Failing Projects		
Project	Interviewee	Quote
2A	Customer Service Employee	“We keep hearing these things. ‘It’s not going to work for them. We don’t have the ability to implement these things.’ We actually had a conversation around how unreceptive to feedback, as an organization we are, and that that mentality needs to start to change. It may have been an influence from her, even just saying it, ‘We need to change how we think about it.’ Then that kind of flipped the switch for me of, ‘Yup, you’re right. That totally makes sense.’ It wasn’t necessarily an action item; it just started to be a mindset of being more receptive to that feedback.”

In her nursing study, Edmondson (2004) found that awareness and acknowledgment of failure enables learning. For cross-functional teams who fail first, there is a two-step process which begins with the awareness that they do not know which is elicited by the failure. The second step and most important component of this process is the acceptance that they do not know. It is only when the team accepts that they do not know that they have achieved they develop a heightened attention to their environment that will support their success. In project 2A, we see how the customer service employee

and her manager become aware of their own embedded thinking, and in order to change this thinking, they must consider the customer in a new way by learning more about them and what they need.

Functional boundary spanning. When a cross-functional team has receptive attention, they exhibit a willingness to cross their functional boundaries of knowledge to learn about other areas of expertise that are not their own. Boundary spanning is an essential communication act in the process of constructing a shared reality. The following examples from interviews present the who, what, where, when, and how of boundary spanning as it occurs.

The where and when of boundary spanning were revealed through interviews as the creation of environments that supported communication. The following excerpts demonstrate how this communication promoting environment is established and used in Table 10.

Cross-functional teams are increasingly geographically distributed, but we found that when teams are in a receptive state, they ensure they create environments and routines that support communication to compensate for these limitations. It is not necessarily the physical location as much as it is the intention to come together in whatever form that can be established to ensure there is an opportunity for exchange.

There are many methods employed for the “how” of boundary spanning employed by cross-functional teams. The team members’ dialogues divulged that they held a very strong sense of their functional separateness and understood its challenges. In their descriptions, they shared the methods they employed to learn from each other despite their separateness. For example, boundary objects (Star & Griesemer, 1989) in

the form of system flow diagrams and process flows were often used as visualization tools to share concepts.

TABLE 10
Environments for Boundary Spanning

Creating Environments for Boundary Spanning and Reality Construction		
Project	Interviewee	Quote
1A	Operations Manager	"We've created the weekly meeting where we just troubleshoot anything and everything every week and still do."
1C	Business Analyst	"Our lead developer was here with us but the three other developers were in India. The fact that one person was here and the other people were there and she could be that liaison was huge. Same with the QA, people were mostly in India but our lead QA was in Charlotte. Then we had our ScrumMaster and myself and then the product owner, and he was the only tough one, he lived in Miami and went back and forth to Mexico City. We just made sure that we did a lot of visual WebEx and all that, so we felt like we were connecting."
1D	Operations Manager	"And everybody in the room, and it was important that it was in the room which we'll get to later on, could actually tick and tie everything back. And you could look in the eyes of someone and believe that the ability to be able to service that." "And actually, they came to the UK for that one, so that they could see... we wanted to do it that way so that they could see the operations that we were in. So, they could walk the floor, see what's there, smell it, touch it, feel it, and know that it's real and it's not the Wizard of Oz, someone sitting behind a big curtain."
1E	Operations Manager	"I kinda got upset, and I said, 'we're never gonna get to what we need until we get all in the room, and understand everybody's part or place in this project.' I finally set up a weekly call, and sometimes even like in a room, we would set up weekly meetings to discuss what did we need to do to get from point A to point B."

In some cross-boundary communication, team members often helped each other to conceptualize their functional worlds without having to understand their intimate detail--demonstrating Boland's concepts of perspective making and taking (Boland & Tenkasi, 1995). Dialogues that best expressed how teams are using boundary spanning methodologies are presented in Table 11.

TABLE 11
Examples of Boundary Spanning Methods

Methods Employed to Span Boundaries		
Project	Interviewee	Quote
2B	Technology Manager	"It's not a document that you create to make somebody happy. You create it so that we know what we are trying to do; it's our blueprint. I mean, would you build a house without a blueprint, you wouldn't."
2D	Project Manager	"When that feedback was given to the consulting firm, they were like okay. It sounds like what we need to do is create one of these roadmaps for each of your line of businesses. That was a major, I think, ah-hah moment for a lot of people because they were like, 'you mean you can't do a one size fits all?' Well, clearly not because everybody had a different idea of what this document would give them."
2H	Business Analyst	"But I think we did strike a balance because he would tell enough detail and be extremely supportive about it too, like took the time to like, 'I need you to understand this because we're gonna do this piece and you need to go do your piece and so here's how it's gonna work and let's work together to now define APIs or whatever, how we're gonna work together.'"
1E	Operations Manager	"And, it was great actually because, his knowledge it was helpful to understand the backend systems without knowing them or seeing them, and understand how they all flowed together."

Accessing knowledge external to the firm. The importance of “who” in boundary spanning was first discovered by Ancona (1990) as an essential factor for teams to achieve success. In her study, she found that teams who accessed external knowledge were successful because of this propensity (Ancona, 1990). Presented in Table 12 are interview examples demonstrating whom the team is accessing with external knowledge in their boundary spanning.

Receptiveness and the team’s awareness that they lacked knowledge was an essential catalyst to access knowledge from outside the team. The objects of reality accessed from outsiders are shared across the project team. The team can then together create new objects of reality within the team as they integrate them with their specific team and organizational context. Their shared acceptance of this new reality’s meaning

becomes their knowledge. In Table 13, we provide excerpts of interviews that demonstrate how team members expressed their experience of a shared reality.

TABLE 12
Accessing Knowledge External to the Firm

Boundary Spanning to Access External Knowledge		
Project	Interviewee	Quote
1A	Operations Manager	"We hired a consultant to help us who understood the payments business, which was Stan. He brought expertise to the table and helped us learn."
1B	Business Analyst	"... people were like...'Oh, yes, we have to design this thing with the people who will be using it.' He designs the first prototype. We start putting it in front of users."
1C	Business Analyst	"Then they learned send Sarah out to go sit with the people who do the job to learn the job."
1E	Operations Manager	"And so... and we used, we had some, firms who were being our guinea pigs, test guinea pigs to see what it would look like, what they think would be best if you know, from a client's aspect like what did they like about the system, the format.""

TABLE 13
Experiencing Shared Reality

Shared Realities Experience		
Project	Interviewee	Quote
2A	Customer Service Manager	"It's in those relationships that people are speaking with you, that they're articulating what their concerns are, they're articulating their perceptions, they're articulating their beliefs. When you start to hear that feedback from your customers and you're getting it directly, you're able to process that information and turn it around and really implement into what you're doing in a very different way."
2A	Product Owner	"What worked in the second iteration better than the first iteration, from my standpoint, was that the advice that we were providing, that we got a shared understanding of what our customers needed."

Discussion

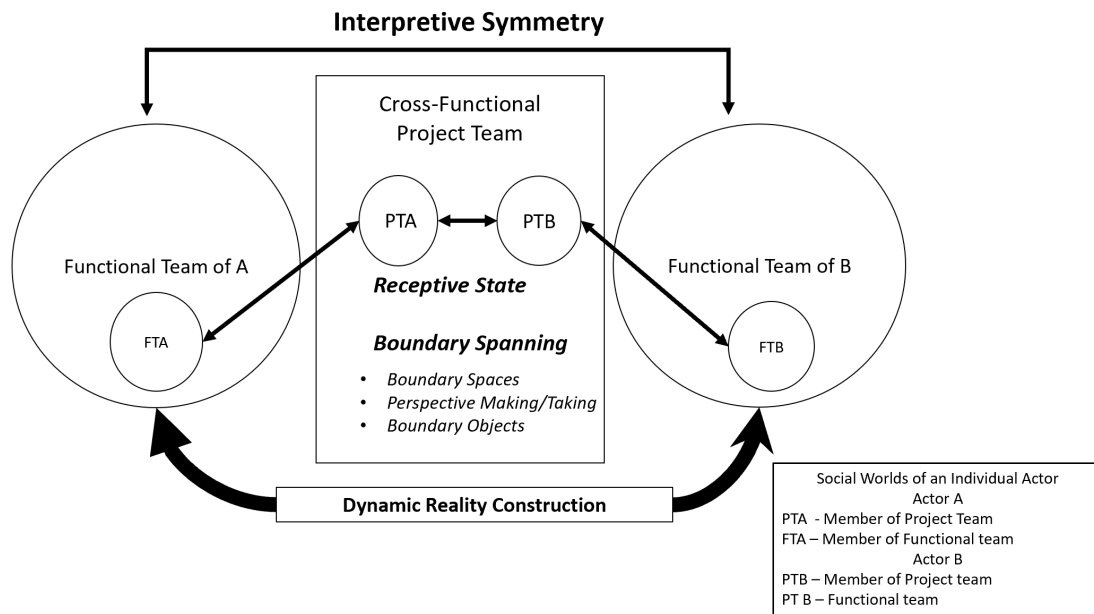
Through analysis of these interviews with cross-functional team members about their project experiences, we have observed the way project teams develop a community consisting of multiple functional worlds which they must bridge and share in the same

reality. We found that their initial state of unknowingness is a critical state that determines whether the project team members can communicate successfully across functional boundaries. The findings from this study demonstrated the means by which the teams established and maintained a shared reality through the processes invoked in boundary spanning. Boundary spanning required that project team members communicate with one another to express their expectations, beliefs, and truths with one another from their functional team's perspectives in order to reconcile them. Without the emergence of a shared reality project teams, members remain trapped in their own separate functional stories that do not adapt.

A process model of interpretive symmetry. A central understanding that emerged from this study was the challenge of communication when the individual team members are rooted in two distinct realities, that of their functional team and that of their project team. Our consideration of communication in this study surfaced how project team members make their functional world known to the project team and the world of the project team known to their functional team. As reality is negotiated across these worlds, the acceptance of what is knowledge must be shared both within the project team as well as within the functional team.

The process model presented in Figure 5 demonstrates how these specific communication behaviors lead to the project and functional teams developing a shared reality across boundaries. We term this process of building a shared reality in multiple social worlds a process of interpretive symmetry, and this model represents the study's contribution to the literature on cross-functional teams.

FIGURE 5
Process Model of Interpretive Symmetry



Given this preeminence of functional identity on projects, we more aptly define the relationships of a group of individuals on a project as an ***alliance***. While members in this project alliance share common goals, we differentiate this structure from a multi-team system such as the simulated air combat teams tested by DeChurch and Marks (2006) which are tightly coupled and contains specialized functions that are highly interdependent and whose work represents a stable configuration of team types to complete the assignment. The functional teams in a project are loosely coupled, joined together for short periods of time, and their degree of interdependence across the teams is subject to the project's goal.

Project alliances are unlike the cross-functional teams studied by diversity researchers where team membership includes individuals with diverse expertise who work within a permanent reporting structure and pursue goals that are familiar to them

(Jehn et al., 1999). For both diversity researchers and team researchers, the primary focus has been on the interactions within the team rather than how the team engages with its external world (Marrone, 2010). Project work requires that individuals interact with others outside their functional team boundaries. Accessing knowledge outside one's functional boundaries can be challenging because functional teams develop their own language and norms related to their expertise that is not shared.

While we are adopting the term cross-boundary, we consider that identity associations with "teamness" as weak or non-existent such that we better define a project as a *cross-boundary alliance*.

We seek to differentiate alignment from the research work surrounding coordination. Team coordination has been studied extensively by researchers within the domain of routines (Bechky, 2006; Bechky & Chung, 2018; Faraj & Xiao, 2006; Kellogg et al., 2006). Coordination is concerned with how routines are formed and how dependencies are managed between activities. Projects differ from routines in that people must come together for a short duration to accomplish an organizational goal (Rose, 2013). More importantly, within a project's context, an individual's role and their relationships to each other and their project goal will exist only for as long as the project does. In effect, when the project ends, so does the world the project team creates that surrounds them.

The process model's configuration demonstrates the essential movement of communication by the actors between their functional team and their project team. The overarching theory which drives the model is the construction of a shared reality, and the social interaction of the functionally disparate actors is central to the development of a

shared reality. Therefore, theories of boundary spanning that support communication and boundary objects and perspective making and taking which support reality construction were identified as the fundamental behaviors in the model. Receptiveness is presented as the affective state that engages the team in the behaviors that create a shared reality among the team members. Shared reality is the cognitive emergent state through which interpretive symmetry is achieved as teams co-create the objects of their shared reality. Their acceptance of this shared reality creates shared knowledge.

The construct of shared reality has provided us with a lens through which we can examine the processes by which a project team interacts or fails to interact with another and gives us a different way to think about why things are unfolding the way they do. Communities sharing the same reality are powerful forces that can accomplish amazing things together through their commitment to communicating with each other and to sharing in their understanding of the task within their functional context which forms their shared reality. The success of the team is defined around the way in which the team is able to maintain a consistent shared reality, as that reality is evolving, so that continued communication across boundaries is what becomes important, and the completion of the task becomes a natural output of a well-constructed and maintained reality.

In summary, we propose that a team's success is dependent on whether they have engaged in the process of sharing in a common reality through the processes of boundary spanning. Their achievement of a shared reality is fluid and requires continuous engagement in a communication process that allows the team to move from the unknown to the known together. The reality sharing process supports them in their ability to acquire and categorize knowledge in a common way, achieving interpretive symmetry.

The task that was set out for them is integrated into the team's reality, and the execution of that task is driven entirely within the reality creating process of the team; the solution becomes a mere output of the reality process. We propose that the true driver in the success of a project is to be found in the team's shared reality.

Limitations

The study's findings are limited by a reliance on an individual's subjective evaluation of whether a project was successful or not. Direct measures of success, such as sales or revenue, provide a more objective means of determining the project's outcome. It is recognized that these measures are often not available, particularly when a project's objective is to serve internal stakeholders. The study was also limited because, although the unit of study was of cross-functional teams, only three projects with a total of nine of the twenty-eight individuals interviewed had more than one participant represented in the study. Having multiple team members is important for corroborating individual perspectives and to search for embedded biases when presenting one's viewpoint. However, because the analysis produced consistent findings both across independent projects and within projects where multiple respondents were interviewed, there is a higher degree of confidence that the findings are consistent.

Implications

In large organizations, efforts to innovate in a rapidly changing world are exclusively managed in the domain of cross-functional teams. The increasing degree of specialization of knowledge is evident throughout society. Is there any one person who can tell you how a watch, a car, or a phone works today? As humans, we are learning how we can cooperate in a way that takes our specialized knowledge and makes it in

some way knowable to others in order to accomplish a goal. Shared reality provides the means by which we can establish bridges across disciplines. In this study, we found that cross-functional teams accessed a shared reality by failure or serendipity. If methods could be developed that move teams to the needed mindful state to access a shared reality, rather than waiting for one or multiple failures, then we have the opportunity to work in a much more effective and successful way in the future.

Future Research

In that the study was open-ended and interviews were not specifically focused on the phenomena of shared reality, this study was limited in its ability to explain more deeply how the reality sharing process is occurring and negotiated on cross-functional teams. Since we believe that this reality sharing is setting up a project for success or failure, it is important to study more deeply the behaviors that are promoting or inhibiting the creation of a shared reality. Berger and Luckman (1991) also extensively discussed the role of identity as an object that is also an integral part of what influences shared reality. When cross-functional team members span the functional boundaries within their own team, each member has an identity that is tied to his/her functional team membership. While much has been researched in the domains of individual and organization identity, less is understood about how identity is navigated across functional team boundaries and its role in a shared reality.

CHAPTER 5: STUDY 2 – QUAN – THE POSITIVE EFFECT OF FUNCTIONAL IDENTITY ON PERFORMANCE

Abstract

Project structures have become a dominating form to organize and accomplish work in firms today. This study focused on firms that source the functional expertise needed to support project implementations on a temporary basis from their organizational hierarchy. In this matrixed structure, individuals reside in two social worlds, that of the project team and that of their functional team. The purpose of this quantitative study was to understand the effect of social identity on cross-functional project teams' communication and their ability to develop a shared reality with other project team members. In this study, we surveyed 212 members of cross-functional project teams in the financial services sector. We found that a strong functional identity was positively related to the ability of an individual to understand what is being communicated from another discipline. The ability to understand someone from another discipline was positively related to a successful project outcome. This study contributes to the literature on team social identity by demonstrating how it is mediated by effective communication across functional boundaries to create successful project outcomes.

Keywords: cross-functional team; social identity; communication; shared reality; performance

Introduction

In a technology-driven, hyper-competitive environment, organizations have become increasingly reliant on the use of project structures to implement rapid change in their processes and products (Söderlund, 2011; Whittington et al., 1999). Projects are

characterized by their limited duration and have a defined beginning and end (Rose, 2013). The temporary nature of projects has led firms to use different organizational strategies to provide resourcing to support their work efforts (Denison et al., 1996). Our purpose here was to focus on communication in projects that are staffed on a temporary basis by individuals from within the organizational hierarchy selected for their functional expertise.

Projects within an organization that are structured and resourced in this temporary way can be thought of as having a heterarchical structure. A heterarchy is characterized by collaboration across a diverse group of experts within an organization with decentralized decision making and accountability (Kellogg et al., 2006). In essence, what we found is that for these projects, the firm is employing a blend of both the hierarchical and heterarchical organizational forms to respond more rapidly to competitive pressures.

Projects have been defined along many different dimensions, but our concern here is the social context of a project (Anbari et al., 2008). Diversity researchers have defined functional experts as individuals with differing informational resources, experience, and social network ties (Argote & Ingram, 2000; Jehn et al., 1999; Mitchell et al., 2015). When a functional expert joins a project, they become a member of a cross-functional team (Denison et al., 1996; McDonough III, 2000). Cross-functional team members on projects are challenged to span two social worlds, that of their project and that of their function.

Cross-functional team members must share their own knowledge and interact and negotiate to create new knowledge together. When knowledge is socially constructed (Berger & Luckmann, 1991) by cross-functional teams, it must be shared and understood

not only among the project membership but also with the functional teams in the organization whose operations are affected by the project (Johnson, Boland, Cola, Aron, & Wang, 2018). Team researchers have noted that differing expertise can create knowledge boundaries (Bechky & Chung, 2018; Edmondson & Harvey, 2018).

Studies to determine whether differing expertise helps or hinders a team performance have proven inconclusive (Mitchell et al., 2015). While some research has found that functional diversity decreases their innovativeness (Ancona & Caldwell, 1992b), other research has found that functional diversity increases team performance (Bell et al., 2011; Jehn et al., 1999).

As members of a functional team, individuals may develop an identity based on their group (Ellemers et al., 1999; Van Der Vegt & Bunderson, 2005). Social identity researchers have sought to explain how identification with a particular group plays a part in whether we successfully interact with others (Lewin, 1948). It has been theoretically argued that social identity and the process of categorization is the source of divisiveness between socially disparate groups (Ashforth & Mael, 1989; Tajfel & Turner, 1986).

Despite this focus on conflict, recent research has found that when cross-functional teams are organized within the same hierarchical reporting relationship on a permanent basis, their functional identity mediated learning to positively affect the team's success (Van Der Vegt & Bunderson, 2005). The project team members in this study have a different organizational orientation with their reporting relationships outside of the project team and the project itself having a limited duration. Given these relational dynamics of the project team members, our research question focused on how functional

identity directly affected the cross-functional project team's ability to learn from one another and perform given these additional variables.

Literature Review

Functional team identity. Tajfel's (1974) work on social identity theorized how individuals bridge their own identity to develop a social one. Social identity theory finds that individuals seek out group membership as a means of supporting their positive self-image (Tajfel, 1974). Social identification represents the degree to which an individual considers themselves to be part of a specific group (Ellemers, De Gilder, & Haslam, 2004). Individuals can belong to multiple groups simultaneously, and the social identity that is invoked by an individual is driven by what they sense is appropriate for the context (Kreiner et al., 2006).

When we develop expertise in a particular field such as technology, we identify ourselves as a technologist. When we work within a group of technologists, there is a mixing of our individual identity as a technologist—"I am a technologist"—with the group—"We are technologists." Our individual identity is shaping our collective identity, and in turn, our collective identity is shaping our individual identity (Ashforth, Rogers, & Corley, 2011). Functional teams will develop a unique social identity that helps them to define what they share in common and supports understanding within their group.

The functional team's identity is not static and must continually adapt. As their environment is changing, they must redefine who they are in relation to those changes (Nag et al., 2007). Identity work requires that individuals are constantly navigating and adapting between their individual and social identity in a process of optimization to maintain coherence (Sveningsson & Alvesson, 2003). Communication, the need for

ongoing dialogue among team members, is crucial to the functional team's ability to maintain a consistent and coherent social identity across their membership.

Communications role in knowledge and identity. We cannot separate who we are from what we know. In the process of creating knowledge, we are also interpreting and understanding our relationship to the knowledge we create through the process of communication (Berger & Luckmann, 1991: 64). This process is not independent, and knowledge and the definitions and categorizations of knowledge and identity are transacted in a social context that is dynamic. We act in our environment, and our environment acts upon us.

Communication clarity. In a cross-functional team context, the ability to clearly articulate one's perspective or to accurately represent the perspectives of one's functional team is essential to the team negotiating an accurate representation of their truth. Berger described the social construction of reality as a process by which individuals engage in the negotiation of their subjective truth through this sharing of knowledge (Berger & Kellner, 1964). This reality is not static and is dynamically constructed, requiring ongoing and continuous validation through communication (Berger & Luckmann, 1991: 29). Establishing an ongoing narrative which tells the story of the evolving reality helps the project team member internalize and retain new shared meaning (Bruner, 1991; Ibarra & Barbulescu, 2010).

The concept of boundary objects has been introduced as a way to support clear communication. Boundary objects can be physical or conceptual representations that support communication across knowledge boundaries (Boland Jr & Tenkasi, 1995; Gal, Yoo, & Boland, 2005; Star & Griesemer, 1989).

Communication frequency. Given that social identity is a transient accomplishment for which the discursive construction and reconstruction is required (Ybema et al., 2009), it is critical for the cross-functional team to maintain communication frequency to sustain a shared coherent context.

Butler (2005) describes this engagement as the process of making accounts which engages identity. She advises that when we try to give an account to another, it will include our own emergent understanding such that we become social theorists, and as such, our story includes our relationship to a set of norms (Butler, 2005). In this dialogue, functional team members must frequently elicit the norms and processes of their teams so that they can be reconciled with members of other disciplines in the creation of their new realities.

Hence, through this process of constructing a reality and making accounts in a cross-functional team environment, members are both intertwining their understanding of self with the larger cross-functional team and reinforcing their own existing functional team identity within those truths. Therefore, we expect when a functional team is strong in their social identity; by necessity, they should also exhibit a high degree of communication to maintain that functional identity.

Theory and Hypotheses

Direct effects of functional team identity. The profession we choose and the expertise we develop is a strong part of our individual identity and our story that defines who we are (Ibarra, 1999). When we share in the same profession and story with others on a functional team, we are reinforcing the identity of ourselves as well as our sense of

membership with the team (Cooper & Thatcher, 2010). We share the same language and understand the nuances of our profession within an organizational context.

When functional team members become members of a project team, their individual and functional identities are called into question (Alvesson et al., 2008). The extensive number of studies conducted on identity threat elucidates the individual's compelling need to protect and defend their separateness through group membership (Petriglieri, 2011). It is important that the functional team member not only protects their sense of self-worth but their functional identity also provides them with a sense of distinctiveness, continuity, and meaning (Vignoles, Regalia, Manzi, Gollledge, & Scabini, 2006). Given that we have established that an individual's functional membership is an essential part of validating their sense of self, we hypothesize the following:

Hypothesis 1a. Functional team identity is negatively related to cross-functional project team performance.

Narrative is essential to the development of social identity (Brown, 2006). Teams with strong functional identity will engage in a continuous dialogue to shape and reshape their identity as their environment changes (Hardy, Lawrence, & Grant, 2005). This continuous use of discourse will be used to articulate functional team identity both within the project team and externally with the functional teams as they seek to maintain alignment during the change that project work is introducing.

This functionally-based discourse supports a clearly articulated functional team context from which others not of that expertise domain can develop a more informed understanding of how the functional team thinks about itself, what it values, and its norms. This enriched expression of thought based in functional values allows cross-functional team members to, in turn, formulate and frame explanations of their own ideas

within their functional counterparts “language.” With this functional context to both give and receive thought, we hypothesize the following:

Hypothesis 1b. Functional team identity is positively related to cross-functional team communication clarity.

Both our identity and our reality are not static but are continually constructed. The frequency with which discourse occurs is essential to maintaining coherence among functional and project team members. Functional teams with strong identities communicate frequently in the process of making accounts to maintain their reality with their own functional team members. Identity is part of what is negotiated dynamically as the functional team members interact not only with their functional team but also with their project team members from other disciplines. Teams with strong functional identities achieve this strong identity through increased communication such that we hypothesize the following:

Hypothesis 1c. Functional team identity is positively related to cross-functional team communication frequency.

When cross-functional teams are able to communicate clearly with one another, they are able to more effectively share ideas and collaborate, constructing a reality that they collectively accept as their knowledge. Individuals with strong identities are already adept at collaborating within their own functional teams. This collaborative mindset transfers from how they interact with their own functional teams to how they interact with their cross-functional team members on the project.

This highly collaborative environment supports the creation of a shared reality among team members. This shared reality enables a more rapid development of knowledge across the team which supports achievement of the project goals. It has long

been established that collaborative efforts that consider external viewpoints produce better outcomes (Ancona & Caldwell, 1992b). Therefore, it is hypothesized that:

Hypothesis 1d. Cross-functional team communication clarity is positively related to cross-functional team performance.

When cross-functional team members increase their frequency of conversation across their knowledge boundaries, they are able to discover more about functional routines, processes, and priorities. Increasing communication also provides cross-functional team members with the opportunity to surface disconnects in their understanding of their reality so that they can create common categorizations of those objects.

Increasing communication also provides the cross-functional team members with additional time to negotiate reality at a more nuanced level. As cross-functional teams achieve a more deeply understood shared reality by communicating more frequently, they are more capable of achieving their project objectives such that we hypothesize the following:

Hypothesis 1e. Cross-functional team communication frequency is positively related to cross-functional team performance.

Mediating role of cross-functional team communication clarity. Crossing knowledge boundaries is the challenge cross-functional teams face. Functional team members with a strong identity have a sensitivity to the need to communicate. When an individual has a strong identity, they have a clear and coherent understanding of who they are and what is important from a functional perspective. This clarity is not only about themselves but also exists in concert with their understanding of their functional team and their individual relationship to that team. The strength of their understanding of these

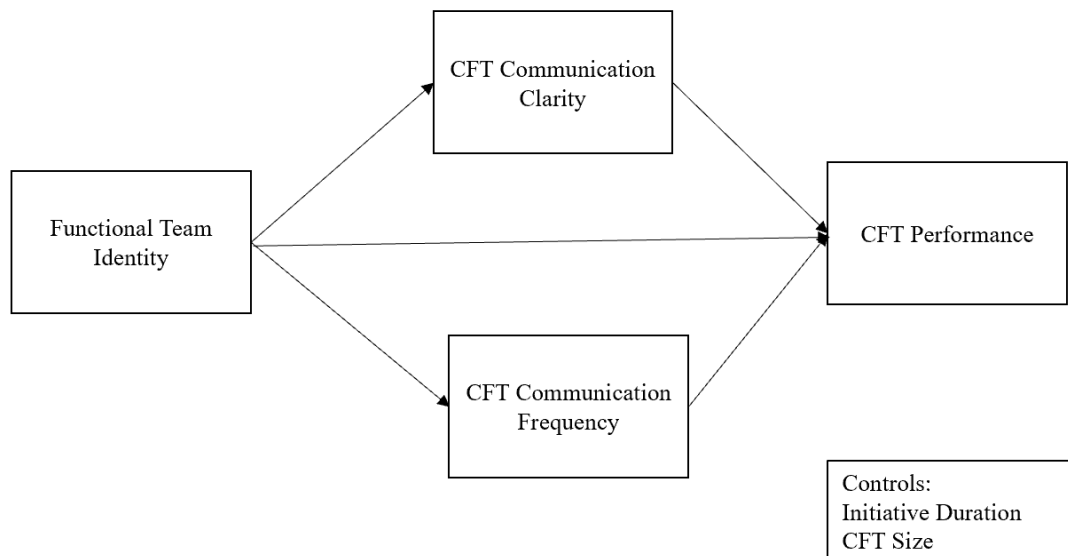
identities includes their routines, their ways of prioritizing and thinking about things in addition to their disciplinary expertise.

The ability to communicate in a way that can be consumed by the cross-functional receiver is essential to spanning the knowledge boundaries between individuals. This allows project team members from other disciplines to easily cross the knowledge boundary and understand the perspective of the functional expert. This supports the creation of a shared reality, and through its acceptance, knowledge. When the cross-functional team shares knowledge, they can more effectively work together to achieve project objectives such that we have hypothesized the following:

Hypothesis 1e. Cross-functional team communication clarity mediates the negative relationship between functional team identity and cross-functional team performance.

We present the hypothesized model in Figure 6.

FIGURE 6
Hypothesized Model



Methods

Sample. The survey was targeted to individuals who had worked on cross-functional project teams in the financial services industry. Project membership was required to consist of three or more disciplines to ensure there was strong cross-functional representation on the project. There were 71 questions on the survey which was distributed using Qualtrics. The author's professional network was engaged via LinkedIn, and the survey was distributed to 950 contacts, of which 311 responded, achieving a 33% response rate. Of the 311 surveys, there were 212 that were fully completed and useable for the analysis, representing 22% of total responses.

Respondents to the survey came from diverse roles within their financial services organization, with 44% of the responses coming from team members with knowledge of specific business functions such as accounting and finance, business management, operations, risk, sales, human resources, and investment management. The balance of the survey respondents was split equally between project management functions at 28% and technology functions at 27%.

The majority of the respondents to the survey would be described as mid-career, with 66% of the respondents between the ages of 35 and 54, 19% of the population below the age of 35, and 15% of the population older than 54. The ratio of men to women was 69% male and 31% female, which is consistent with the lower representation of women in both financial services and in technology fields. Tenure with the firm was divided fairly evenly between those with a long tenure of five years or more representing 42% of the respondents, with the remainder of the respondents having worked for their firm less

than five years. The respondents were primarily from North America at 88% with the balance of the responses received from other countries.

The locations of the cross-functional teams were diverse, with co-located teams representing 46% of the initiatives, while virtual teams were 13%, and a mixed team of both virtual and co-located teams was 41%. We wanted to understand at what point in the change initiative's life cycle the respondent was reporting from when completing the survey. Results showed that 32% of respondents were within the first nine months of the initiative, 31% were between nine and eighteen months, 28% were eighteen months or longer, and 9% of the initiatives were responding about a completed initiative. Of the total initiatives, 50% had twenty team members or less, 27% had twenty to fifty team members, and 22% had greater than fifty team members participating.

Measures.

Functional team identity. Functional team identity was defined by referencing the work of Tajfel (1974), who first expressed the concept that we think of ourselves not only as individuals but also as members of groups, and this group membership represents a social identity. In her research, Ellemers found that there are three distinct dimensions of social identity: affective, evaluative, and cognitive (Ellemers et al., 1999). For the purpose of this study, we chose to use the affective component of social identity as our measure which captures how an individual feels about their team members, the emotional component of social identity.

The scale for the affective component of social identity was adapted from Cameron (2004) to place it in the context of a functional team in a business environment and included four of the six items from his original scale. The respondents were asked to

first think about their functional team members with whom they shared the same discipline and then respond to the following questions: (1) I felt we had strong ties; (2) I really “fit in” with them; (3) I belonged with them; and (4) I had a sense of connection with them. Response options ranged from 1 = “strongly agree” to 5 = “strongly disagree.”

Cross-functional team communication clarity. Communication clarity was adapted from a scale originally used to evaluate teacher performance and is defined as the ability of a teacher to communicate information in a way that can be clearly understood (Chesebro & McCroskey, 1998). The communication clarity scale was adapted to measure how well the respondent from the functional team was able to understand communication from other cross-functional team members. The survey questions were introduced with the following statement: “When I had questions or did not understand something about the project I found that....” Then, the individual was asked to answer the following questions: (1) my questions were answered clearly; (2) explanations provided were generally straightforward; (3) new concepts I was not familiar with would be clearly defined; (4) relevant examples would be used to help clarify; and (5) generally, I could understand what was being explained. Response options ranged from 1 = “strongly agree” to 5 = “strongly disagree.”

Cross-functional team communication frequency. The communication frequency scale contained three items and was adapted from Kacmar, Witt, Zivnuska, and Gully’s (2003) original 8-item scale that assessed work-related communication frequency on employee performance. The definition of communication frequency in this study is measuring all communication with a cross-functional team member regardless of its purpose. The questions were preceded with the following statement: “With cross-

functional team members, how frequently did you communicate...?” and the following questions were asked: (1) by email? (2) by chat? and (3) verbally? Response options ranged from 1 = “more than once a day” to 5 = “less than once a month.”

Cross-functional team performance. Performance is defined as the ability of the cross-functional team to satisfy customer needs and expectations, and the scale was adapted from a self-assessment measure published by Edmondson (1999), of which four of the five items from this scale were selected for use in the study.

The measures were preceded by the following statement: “When working on the project...” and the following questions were asked: (1) Those who received or used the work of the project team often had complaints about it; (2) The quality of work provided by the project team did not improve over time; (3) Critical quality errors occurred frequently on the project; and (4) Others in the company who interacted with the project team often complained about how it functions. Response options ranged from 1 = “strongly agree” to 5 = “strongly disagree.”

Controls. Cross-functional teams vary in size, and prior research has shown that there is a relationship between team size and both cohesiveness and internal communication, we included team size as a control variable (Ancona & Caldwell, 1992b).

The point in time at which the cross-functional team members had been working together on their initiative varied by respondent. This analysis includes a control variable to measure duration given that prior research has shown that the length of time a team works together affects their performance (Harrison, Mohammed, McGrath, Florey, & Vanderstoep, 2003).

Measurement model.

Exploratory factor analysis. This analysis was prepared using responses from 319 surveys, for which 111 were removed due to incomplete responses. Within the survey data, there were 11 respondents with one missing data element for which the mean response value for that respondent was used to replace the missing values. Our final sample size for the data set was 218.

There were a total of five factors and 20 items represented in brackets in the analysis; the affective component of functional team social identity (5), communication clarity (4), communication frequency (3), performance (4), and a measure to test for social desirability (4). The items were tested for normality, and both skewness and kurtosis for all items were below the acceptable thresholds (West, Finch, & Curran, 1995). The performance measure was tested for outliers, and none were found.

The exploratory factor analysis was conducted using SPSS with Maximum Likelihood and Promax for the oblique rotation (Matsunaga, 2010). The initial load showed SocDes01 and Perform04 with negative eigenvalues in the pattern matrix, so we removed these items to achieve our final solution with 18 items. The final solution achieved a KMO measure of .827. The table of reproduced correlations had 15 or 4% non-redundant residuals. Factor loadings for all items were greater than .32 threshold (Tabachnick & Fidell, 2001). The Cronbach's alpha for our final model was .86.

Confirmatory factor analysis. The confirmatory factor analysis was conducted using Amos using Maximum Likelihood. The model had good fit statistics with the following results, CMIN/DF= 01.72, CFI = 0.92, RMSEA= 0.06, PClose = .137, and SRMR=.07 which were all within acceptable ranges of fit (Hu & Bentler, 1999). The

model achieved convergent validity and composite reliability with AVE above the 0.5 and the reliability measures exceeding the threshold of 0.7 (MacKenzie, Podsakoff, & Podsakoff, 2011).

The model was examined for common method bias by using social desirability as our marker variable. A chi-squared difference test was conducted to compare two alternate nested models. For the first model, regression weights were set equal to each other, and for the second model, weights were set to zero. The p-values for these models' differences, when compared with our default model, were greater than .05, indicating the lack of common method bias. Composites were then created in Amos, excluding our social desirability factor.

The composite scores were used to test for outliers using Cook's D, and there were no values greater than 1.0 (Chatterjee, Hadi, & Price, 2000). Testing for multicollinearity, the variance inflation factor (VIF) was below a threshold of 3 for the dependent variable performance (Kock & Lynn, 2012).

Structural equation model. We used Mplus to create our structural equation model. The model had good fit statistics with the following results, CMIN/DF= 01.73, CFI = 0.99, RMSEA= 0.06, PClose =0.33 and SRMR=0.03, which were all within acceptable ranges of fit (Hu & Bentler, 1999). Table 14 shows the means, standard deviations, and correlations with Cronbach's alpha listed on the diagonal.

TABLE 14
Descriptive Statistics and Correlations

	Variable	Mean	s.d.	1	2	3	4
1	Performance (CFT)	2.12	0.99	0.91			
2	Functional Identity	1.88	0.73	-0.05	0.86		
3	Communication clarity (CFT)	2.37	0.68	.68***	0.64***	0.80	
4	Communication frequency (CFT)	1.45	0.59	-0.15	0.30***	--	0.86

n=212

***p< .001

Cross Fuctional Team (CFT)

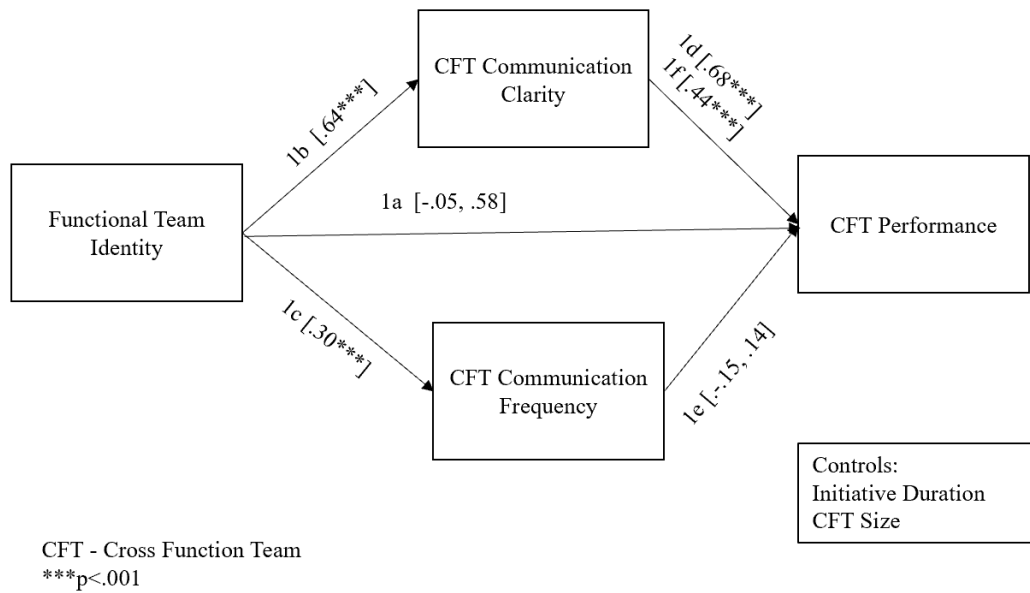
Cronbach's alpha are presented on the diagonal in bold.

The squared multiple correlations for each structural equation were as follows: CFT communication clarity $R^2 = .403$, CFT communication frequency $R^2 = .16$, CFT performance $R^2 = .245$.

Findings

Hypothesis 1a was supported, showing that there was a negative relationship, but it was not significant between functional team identity and cross-functional team performance, $\beta = -0.05$, $p = .588$. Hypothesis 1b was supported and significant with functional team identity, demonstrating a positive relationship to cross-functional team communication clarity, $\beta = 0.64$, $P < .001$. Hypothesis 1c was supported with functional team identity positively related to cross-functional team communication frequency and significant $\beta = .30$, $p < .001$. Hypothesis 1d was supported and significant with cross-functional team communication clarity positively related to cross-functional team performance, $\beta = 0.68$, $p < .001$. Hypothesis 1e was not supported and not significant, cross-functional team communication frequency was negatively related to cross-functional team performance, $\beta = -0.15$, $p = .14$. The results of the direct effects are presented in Figure 7.

FIGURE 7
MPlus Results for Hypothesized Model



Mediation effects. The mediation analysis was performed using the Hayes and Preacher (2013) method with bootstrapping. The bootstrapping samples were set to 2,000, and the bias-corrected confidence interval was set to 95 before executing the model. Hypothesis 1e was supported; the results showed cross-functional team communication clarity fully mediated the positive relationship between functional team identity and cross-functional team performance $\beta=0.44$, $p<.01$, CI [0.315,0.565]. Results are presented in Table 15.

TABLE 15
Mediation Effects of Communication Clarity on Functional Identity

Variable	Direct Effect	Indirect Effect	Total Effect	Confidence Interval (.05)	
Functional Identity	.44***	-0.05	.44***	.315(lower)	.565 (upper)

*** P <.001

Controls. The effect of project team size on performance was $\beta=0.08$, $p=.26$, which was not significant. The effect of project duration on performance was $\beta=.00$, $p=.96$, which also was not significant.

Discussion

This study sought to better understand how one team's functional identity affected the ability of a cross-functional team to communicate with one another and how this, in turn, affected cross-functional team performance. While previous research has considered the development of the social identity formed within a multi-disciplinary team and how that positively effected multi-disciplinary team performance (Van Der Vegt & Bunderson, 2005), we have no examples of studies seeking to understand the mixed effect of how one team's identity might affect another.

Our first hypothesis, which proposed that functional team identity would negatively impact team performance, was not supported, and there was only a small negative effect that was not significant. In and of itself, functional identity does not support nor detract from the team's performance. Nevertheless, in the presence of learning about another's functional identity, we see a strengthening of the team's ability to perform. When members of project teams are able to learn about each other's differences, considering those differences becomes an essential part of what makes the team successful.

Communication clarity and frequency results showed that cross-functional exchanges with partners play a role in strengthening identity when these interactions occur. We consider that when someone receives knowledge from across a boundary and when there is effective communication, the receiver will be able to articulate questions in

response. These questions in the form of communication frequency help the receiver understand how the information that is shared from another discipline relates to the sender's own discipline.

The receiver is, in turn, integrating this knowledge with their own understanding of their functional team, increasing functional identity. The context provided by the sender comes from representations about the sender's functional team, relaying how that team thinks about their processes and procedures and what they have decided is important and why. This finding provides insight into the interrelationship between boundary communication and identity, but only the clarity of communication has an effect on team performance.

This study demonstrated how communication acts as the bridge that facilitates the expression of identity and how the expression of those contextual aspects of a team's functional identity are necessary to achieve cross-functional team performance. Gal et al. (2005) theorized that boundary objects provide the mechanism by which a functional identity can be communicated to members from other disciplines. However, to our knowledge, this is the first quantitative study that directly measures how a team's functional identity affects the larger cross-functional team.

Central to the premise of this study was the concept of a socially constructed reality that is subjective and requires a continuous dialogue of making accounts with one another to support the dynamic nature of how functional teams are defining themselves over time. From these results, we learned that functional team identity can positively influence the performance of a cross-functional team, and the way this is achieved is through communication clarity.

Functional teams with a strong identity have a need to continually communicate to validate who they are, and part of this continuous validation is occurring with members of other functional disciplines. Through the use of boundary objects, the cross-functional team members are able to establish clear communications across disparate functional knowledge, which leads to positive performance outcomes.

While we have established a relational link on why teams with a strong functional identity will need to drive strong communication efforts when engaged on a cross-functional team, scholars have pointed out that there is a lack of integration between research in social identity theory and research in communication as called out by researchers from both fields (Hogg & Reid, 2006; Scott, 2007). The results of this study put forward some initial findings that demonstrate how communication and social identity interact.

We have all been subjected to the experience at one time or another of thinking that we are on the same page with someone and finding when too much water has passed under the bridge that we were never on the same page at all. Considering the fluidity of our reality and our self-definitions of who we are, which includes our group identity of the functional team we are a member of, we are required to continually keep an account with one another. The expression of our social identity is integral to and becomes embedded in the interpretation of the knowledge we share with others. We hope that the findings of this study help to shed some additional light on the dynamics of how cross-functional teams can work more effectively together.

Limitations and Future Research

While past research and theory on boundary objects and their relationship to social identity was employed to explain how functional team identity establishes cross-functional team communication clarity, this study did not capture information about the use of boundary objects.

As discussed throughout the study, the process of constructing reality is incredibly dynamic, particularly for teams engaged on temporary projects. The study's methodology constrained the analysis to considering and measuring the factors in a linear process rather than a recursive process. This limitation presents an opportunity for future studies to consider alternate methods to capture the dynamic aspect of the reality construction process.

The measurement of cross-functional team performance reflects the subjective view of the respondent, which can be subject to bias. This is generally a limitation in the study of teams for whom performance measures are not captured in an objective measurement system.

Given that the design of our study was cross-sectional, there are alternate explanations for the positive team performance, which were not part of the study. Designing a study that occurs across time would help to further validate these results.

Practical Implications

This study provides management with insight into the dynamics of how a positive functional team identity can have a beneficial effect on cross-functional team performance. Managers should expect that initiatives that have functional teams with strong social identities will perform better than when a functional team's identity is weak.

We understand that strong identities require constant validation through continuous communication.

Monitoring cross-functional team communication to gauge when it may be declining or, in general, lacking can provide early indications that functional identities are weakening or are weak reducing cross-functional team communication clarity. The weakening of frequency of communication across functional disciplines, particularly during the early discovery phase of a new change initiative represents a significant concern. It is important for both the functional team and the cross-functional teams to develop and continually evolve a shared reality. In the absence of frequent cross-functional team communication, clearly understanding one another becomes a challenge which, in turn, will lead to a decline in the overall performance of the cross-functional team.

Conclusion

We have all been subjected to the experience at one time or another of thinking that we are on the same page with someone and finding when too much water has passed under the bridge that we were never on the same page at all. Considering the fluidity of our reality and our self-definitions of who we are, which includes our group identity of the functional team we are a member of, we are required to continually keep an account with one another. The expression of our social identity is integral to and becomes embedded in the interpretation of the knowledge we share with others. We hope that the findings of this study help to shed some additional light on the dynamics of how cross-functional teams can work more effectively together.

CHAPTER 6: QUAL – ALIGNMENT OF PROJECT GOALS WITH FIRM PURPOSE – EFFECTS ON FUNCTIONAL IDENTITY AND PROJECT PERFORMANCE

Abstract

For a project to be successful, team members must express their functional identity with one another across their social and knowledge boundaries. The purpose of this study was to understand how a cross-functional project team communicates functional identity in their process of constructing a shared reality. We use shared reality theory and a qualitative approach to analyze the video and audio recorded meetings of three cross-functional project teams over a four-month period in a mid-sized private fintech firm.

We found that teams with a shared reality readily accepted each other's communication about functional identity across boundaries. We found that in the absence of a shared reality project, team members questioned both each other's functional identity and their project team's reality. We discovered that the project goal played a central role in supporting the construction of a shared reality among the project team.

Project goals serve as a representation of the firm's purpose. Functional teams have an established relationship through their identity that supports firm purpose. When a goal does not appropriately represent firm purpose the relational items of identity between functional team and firm are disrupted. This study contributes to the literature on communication and identity and their effects on performance. It also contributes to the literature on shared goals by identifying the antecedents that support their creation. The findings of this study also provide empirical support to the theory of a shared reality.

Keywords: shared reality; shared goals; cross-functional teams; identity

Introduction

Businesses, as a matter of course, continue to invest in creating new capabilities that will help customers solve their problems, and organizations execute much of this work through project teams. Project teams often struggle with communication and achieving alignment on their goals (PMI, 2013). In this study, we sought to explore the communication of functional identity in project team interactions given identity's role in project team performance. We also sought to understand how team members related to the project goal within the context of functional identity given that shared goals have been shown to also positively effect team performance (Carson et al., 2007)

Project goals help firms to achieve their strategic objectives and therefore are an extension of the firm's purpose. Functional team members have an existing relationship with the firm that is interrelated with their identity. There is an implicit relationship between a functional team member's identity and project goals as these goals are a representation of firm purpose. Functional team members are aligned with project goals through their identification with firm purpose. Functional team members must also align with each other across their social boundaries within the project.

We seek to differentiate alignment from the research work surrounding coordination. Team coordination has been studied extensively by researchers within the domain of routines (Bechky, 2006; Bechky & Chung, 2018; Faraj & Xiao, 2006; Kellogg et al., 2006). Coordination is concerned with how routines are formed and how dependencies are managed between activities. Projects differ from routines in that people must come together for a short duration to accomplish an organizational goal (Rose, 2013). More importantly, within a project's context, an individual's role and their

relationships to each other and their project goal will exist only for as long as the project does. In effect, when the project ends, so does the world the project team creates that surrounds them.

We considered the social construction of reality as an appropriate theory to gain an understanding of how these newly formed teams engage in the process of constructing their relationships within the context of the project. We find that the project's goal is the central point on which all relevant relationships engaged in the project must align. These relationships will inform the project goal or be touched by the project's goal.

In the literature, theory relating to firm purpose has been conceptualized as organizational identity (Albert, Ashforth, & Dutton, 2000). In the sense of the firm as having its own being, we propose that the project goal exists as a representation of the priorities, wants, and needs of the firm itself. We can see that the project goal relates to a firm in the same way an individual's goal relates to the individual. Goals are motivated by the unique needs and desires of the individual or organization, and we can know something about the person or the firm by examining their goals (Brown, 2015). Therefore, we argue a project goal must be a rational and consistent representation of the firm's purpose in order for it to be believable. Shared beliefs are central to how a shared reality is established (Echterhoff et al., 2009).

In this study of three project teams, we analyzed the recorded team meetings of a Fintech firm which occurred over a period of three months. All project participants had to confront a significant amount of uncertainty as they face unexpected events from one team meeting to the next. We present empirical evidence demonstrating how a shared reality is created or is not created among a project team based on how the initial project

goal is defined and understood. We show how this shared reality supports a team's ability to navigate these unexpected events by responding adaptively. Teams without a shared reality are driven to interpretive loops to establish meaning with each event. These teams continually question the motives, roles, and actions of their teammates and partners. What, in effect, these teams have is a reality that is constantly being called into question.

Literature Review

Shared reality. What distinguishes humans from other animals is the unique, complex and varied ways in which we communicate meaning as has been explored by Wittgenstein (1974) in his analysis of our uses of language. Through communication, meaning is not achieved directly but is a social process, and meaning is negotiated through one's sharing in the construction of reality with other humans (Berger & Kellner, 1964; Berger & Luckmann, 1991; Bruner, 1991; Echterhoff et al., 2009). Berger and Luckmann (1991) defined reality from a sociology perspective as a phenomenon having an existence that is separate from our own and that, in turn, knowledge is the belief that the phenomenon is true. Echterhoff defined the term reality from the perspective of psychology as one's subjective experience that something is true, and he notes that this truth exists for an individual regardless of its verifiability (Echterhoff et al., 2009). We, therefore, use these ideas of a specific object or phenomena which is perceived as truth as the meaning of reality.

When an object has the same meaning for a given group, then that object will be seen in the same way by that group (Blumer, 1986). The beliefs and truths or our reality are importantly connected with other members of the group (Bar-Tal, 2000). We adopt the definition of "shared" in the term shared reality as beliefs or feelings that are held in

common among a specific group (Echterhoff et al., 2009). Whether or not a group is able to form a shared reality is dependent on the process of communication they employ.

Vygotsky's (1962) key insight on social interaction is that we use both language and symbolic systems to shape the reality we present to one another. While reality is shaped by the systems of communication we rely on, it is also shaped by the individuals with whom we must communicate. The extent to which we shape reality by considering the person we are engaged in a dialogue with determines how successfully we can achieve a common interpretation (Higgins, 1992). The consideration of another's view includes what the other's expectations may be regarding how individuals in the group perform their roles (Higgins, 1992). The consideration of another person is often referred to as stepping into another's shoes. Importantly, Mead notes that through this enactment of considering another, we can establish that there is a self which is separate and distinct (Blumer, 1986).

By considering ourselves and the other in our representation and negotiation of reality, we are, in essence, binding that individual or group of individuals and ourselves in relationship to that object of reality and to each other. This interlinking of the individual and the object in our experience creates an opacity such that "Identity remains unintelligible unless it is situated in a world" (Berger & Luckmann, 1991: 174). Therefore, it is through our objects of reality that we find not only meaning for the object but through which we can come to know our own self in relation to both the object and to others.

In the process of our daily existence, we continuously encounter unexpected events that in some way affect our negotiated reality and may give reason for us to

examine it. We refer to the extensive literature on routines that acknowledge that we have an innate sense of urgency to make sense of our world and rectify inconsistencies (Dionysiou & Tsoukas, 2013; Feldman & Rafaeli, 2002; Weick et al., 2005). From the perspective of a shared reality, these events expose not only our objects of reality but our identity in relation to those objects.

The argument we wish to make here is that whether a shared reality has been successfully constructed or not will determine how well a group will be able to navigate these events of unknowingness together. We expect that groups who have established a shared reality can react adaptively without the need to reexamine and verbalize their understanding of the object of reality in question nor their relationship to it. That when things make sense, their understandings can remain below the surface, and they exhibit a “roll with the punches” response to change, adapting their understanding and their acts dynamically and in concert with each other as needed. Alternatively, we expect that if a shared reality has not been established about the object of reality, then each new event will call into question the meaning of the object and the group’s relationship to that object of reality.

Having established what shared reality is and the requirements of how it is formed and the outcomes of whether it is established or not, we now consider the explicit conditions of constructing a shared reality. Echterhoff prescribed four conditions that can tell us whether a shared reality exists (Echterhoff et al., 2009). He prescribed that there must be an object of reality for which everyone shares a common inner state defined as one’s beliefs or feelings about that object (Higgins & Pittman, 2008). Additionally, an individual must be able to have an awareness of the other’s belief that they share in this

belief and that the motivation which underlies this reality is inseparable from the reality itself (Echterhoff et al., 2009).

We have taken these four conditions and applied them to our cross-functional project team setting as follows: (1) the object (Berger & Luckmann, 1991) of reality that is shared among the cross-functional team is the project goal; (2) the team shares a common belief about the goal and they are aware of it; (3) the purpose that motivates the goal is central to the goal as the reality object; (4) shared reality cannot exist unless the teams beliefs are connected in relationship to the goal.

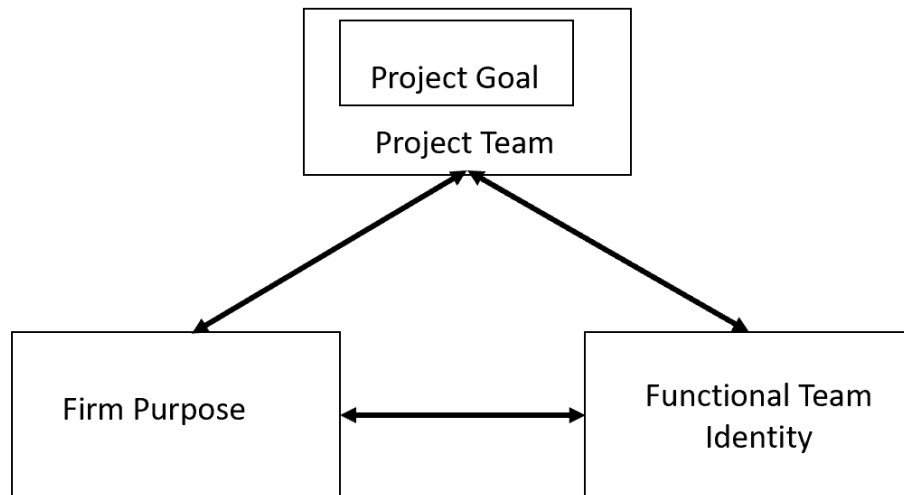
Shared goal as an object of reality. Individuals develop actions in relationship to the objects of their reality; therefore, in order to understand the actions of individuals or groups, it is important to understand the objects of their reality (Blumer, 1986). As such, we consider the project goal and its meaning as the central object of reality for the individual members of the cross-functional project team. We rely on the team's actions to understand what meaning they have provided to the goal.

Past research on team goals has demonstrated that a shared goal leads to project team performance (Carson et al., 2007). A limitation of relying on this past study on shared goals for our purposes was that it was conducted using consulting teams which represented an inter-team context. Our project teams are teams of teams and as such represent as intra-team construct. Team members in an intra-team communication environment have social and knowledge boundaries which challenge their ability to communicate in order to establish shared goals.

We propose that a goal can be thought of as situated in its own firm-derived context and as an object of reality, consists not only of the end which is being strived for

with that goal or the “what,” but it must also be motivated by reasons of “why” which is essential to the goals meaning. Here, note that when team members are crossing functional boundaries, we assume the object of the goal is standing independently as an object with its own reality, its own “why.” Hence, when team members are crossing knowledge and social boundaries to establish a shared goal, we are proposing a new way in which we can think about how that goal is shared and how individuals are enacting their relationship to it. We present these relationships between functional identity, firm purpose and project goal in Figure 8.

FIGURE 8
Functional Identity Relationships



Research Design

Methodology. In this study of three project teams, we analyzed the recorded team meetings of a Fintech firm which occurred over a period of three months. All project participants had to confront a significant amount of uncertainty as they face unexpected events from one team meeting to the next. We present empirical evidence demonstrating how a shared reality is created or is not created among a project team based on how the

initial project goal is defined and understood. We show how this shared reality supports a team's ability to navigate these unexpected events by responding adaptively. Teams without a shared reality are driven to interpretive loops to establish shared beliefs with each event. These teams continually question motives, roles, and actions of their teammates and partners. What, in effect, these teams have is a reality that is constantly being called into question.

Our approach to the analysis was based in the theory of socially constructed reality, and we sought to capture statements that related to an individual's beliefs and acts in the context of the project. These beliefs and acts are reflections of each individual participant's relationship to the object of their reality, the project goal. Following a similar analytical approach to grounded theory but employing our boundary conditions, we began to organize these statements into lower-level themes and then reduced them to higher-level themes (Charmaz, 2014).

We maintained a separate coding structure for each project in order to be able to separate the themes within each project and then be able to compare those themes across projects. We also organized the coding across time and project to understand the relationships among the codes across time and in order to allow again for the comparison with each project.

In that shared reality engages the self in the construction of the object of reality, we employed separately a survey on identity. We reasoned that when a team has developed a shared reality, they will have a stronger sense of identity as measured by their emotional bond with their functional team, their project team, and in relationship to their organization.

Sample. Approval to conduct the study was secured from the CEO of the company, and a non-disclosure agreement was executed on behalf of the research team. The primary researcher worked with the head of the Project Management Office to enlist projects for the study. The request was for four to five projects with a mix of low and high performing projects as evaluated by the PMO leader in consultation with a senior organizational leader on business implementations. Project team members were required to consent to the study, and the researcher secured agreement from all members to participate in the study.

There were initially four projects selected for the study: two which were considered to be successful and two that were not successful. The two unsuccessful projects were starting a reset after a prior failed implementation. The initial expectation was to collect five video recorded project team meetings from each of the project teams. Within the first month of the study, the manager for one of the successful projects left on paternity leave and then subsequently left the company resulting in this project being dropped from the study. It was decided to increase the project study from a three-month to a four-month analysis and capture seven recordings from each team meeting rather than the five that had originally been planned. We present a summary of the projects in Table 16.

TABLE 16
Project Descriptions

Project Number	Project Description	Total Team *	Business Functions Represented	Project Stage	Meeting Cadence	Meetings Recorded
URP	Implement a utility payment solution at one of the divisions of an existing client.	8	Project Management, Product Management, Sales, Technology, Payment Operations, Customer Care	Restarting – 2 Prior Failures to Implement	Weekly	8
BP	Add new features to the existing payment application.	12	Project Management, Product Management, Technology, Architecture	Restarting – 1 Prior Failure to Implement	Weekly	8
CP	Implementing the existing payment solution through an international partnership.	19	Project Management, Technology, Payment Operations	Ongoing	Weekly	7
* The team number represents the actual number of members who directly participated in the recorded meetings that were analyzed. Some individuals either never participated or subsequently left the company.						

Data collection. The initial permissions from the participants were collected between October and November of 2019, at which time recordings began. The researcher met with the project managers (PM) via phone and took notes about the project's history and purpose as it was known by the PM. Two of the PMs had recently been assigned to the CP and URP projects, respectively, while the third PM was continuing in a second reset of the BP project. Teams were expected to arrange meetings in conference rooms where video recording capabilities were present. One of the teams was not successful with the video recording but provided audio recordings. Agendas for the meetings, when prepared, were accessible for the meetings that were video recorded. Although team meetings were expected to occur weekly, meetings tended to be sporadic due to the holiday season in November, December, and early January or due to other conflicting events scheduled within the organization. Collection of recordings continued into the beginning of February, totaling seven video or audio recorded meetings for each project.

A Qualtrics survey was used to document team members' agreement to participate and to also administer a short survey to assess each team member's level of identity within the organization, the project team and their own functional team. The

identity scales and the reference source for the scale used as indicators for collective team identity are presented in Table 17. We relied on the affective scale of social and organizational identity as a measure of the team's emotional connection to their team members without the cognitive and evaluative scales. We administered the survey using a five-point Likert scale in our measure. Due to the small sample size, we used identity purely as a reference indicator and created an average from the entire team's responses.

TABLE 17
Identity Measures

	Functional Team Identity	Organizational Identity	Project Team Identity
Introductory Statement	When I think about my <u>coworkers in my department</u> who worked in the same business function with me...	When I think about my company...	When I think about the <u>project team members</u> who work on this project with me, I feel that...
Q1	I really "fit in" with them.	I feel like 'part of the family'.	I really "fit in" with them.
Q2	I belong with them.	I feel 'emotionally attached'	I belong with them.
Q3	I have a sense of connection with them.	I feel a strong sense of belonging to my company.	I have a sense of connection with them.
Scale Source	(Cameron, 2004)	(Allen & Meyer, 1990)	(Cameron, 2004)

Data analysis. Team meetings were transcribed using NVivo Transcription to protect company privacy. The transcriptions were then imported into NVivo software and the researcher listened to the recordings and corrected any errors in these electronically transcribed meetings. As the transcriptions were received, the researcher began coding and then organized the coding by themes. Early coding and themes informed coding in later transcriptions. This coding went through a second level of consolidation into higher-level themes when all coding was complete.

The coding organization was also structured to differentiate each project from the other to observe differences in patterns between projects. A summary of the coding

across the three projects was created in Excel so that differences in themes could be more easily observed. As we uncovered the differences in themes, we began the mapping of performing and non-performing projects based on these themes across time. This process provided us with insight into the temporal relationships within the data that was distinct to each project team.

In addition to the qualitative analysis of the interviews, we began the project with an interview with the PM which was not recorded but notes were captured and included with the NVivo data. We used this material as a reference for how the project was defined and to provide context on the history for consideration in the final analysis. We also included in this study a brief identity survey at the beginning and end of the study to gain a sense of the team's level of identification with their functional team, project team and organization. We expected that these results could provide additional insight into the team's level of commitment and engagement with the project and whether events within the project influenced their feelings over the time period studied. We combined the team's responses in a spreadsheet and used an average to compare identity between the teams and across time.

Findings

Measures of shared reality. We refer back to our earlier premise that while we cannot directly observe an object of shared reality, we can come to know it through observing the acts of the group in which it is represented. Through our coding and analysis, we found that the CP team had a shared reality while the BP and URP teams did not. This was evidenced by whether there was agreement between functional team members about functional identity and reality.

We had previously argued that moments of unknowingness would affect how a team reacts depending on whether their reality is shared or unshared. We, therefore, sought to capture in our coding the number of events of unknowingness the team experienced. The coding definitions as they related to expressions of identity and reality are presented in Table 18, with results presented in Table 19.

TABLE 18
Coding Definitions

Code	Code Definition	Example Statements
Identity	An individuals statement about their own identity or statements they made about their functional team, project team, organization or external parties identity. This also includes statements from others who affirm the identity by agreeing with the statement.	"So when I asked Aniket, I said, "when are you going to be done?" He said, "well, we've got to work through and get agreement cetera." So the project management style of that would be the person who's building the widget. When are you going to be done? Give me a date. I still haven't got that from Aniket" BP PM
Interpretive Symmetry	Acts either in the past, in the present or planned for the future that seek to expose functional knowledge and align it cross-functionally among team members and external constituents.	"Let's verify with Brian that , you know what we're doing matches . But we should also match up against the processes that we outlined when we designed all this . If we're doing something different , either adjust what we're doing , or adjust the process flow to match what we're doing ." CP Outgoing
Projected Reality	Statements that involve the engagement of negotiating a future reality. These statements start with an individuals interpretation of people or events which leads to a proposed plan of action to respond. These statements include their reasoning and justification for that proposal and the associated challenges and counterproposals others make in	"If the South is interested let's choose a supplier or 10 accounts or whatever it is and let's just get that small tiny little pilot volume on and prove it out and give you something to touch and feel before we go into a full implementation with any markets. Thats the only thing I can think of doing" URP Product Manager
Process	References to existing processes used in the organization or the awareness that a new process is needed.	"Yea, if they're getting payment emails that's fine . And if we're sending them welcome e-mails or doing any sort of marketing which marketing is like , no , thank you Canada , we don't want you to touch it." CP Supplier Relations
Unknowingness	Unexpected events which occur. Instances where the person with the knowledge is unavailable to the project team to provide understanding or it is not understood who should performing a particular role or where the information can be found.	"So and that's what I'm saying. So at this point we have driven the curvy road, we've gone up, we've gone down and you just keep coming with requests and even when we folded and we said OK we'll do this pilot, that still wasn't enough. So what's the purpose.." URP PM

TABLE 19
Summarized Coding Analysis of Project Team Meetings

Coded Statements	BP	%	URP	%	CP	%
Identity	51	17%	49	14%	17	8%
Interpretive Symmetry	50	16%	45	13%	61	30%
Projected Reality	133	44%	194	56%	52	25%
Process	19	6%	25	7%	28	14%
Unknowingness	51	17%	33	10%	47	23%
Total Project Coding	304	100%	346	100%	205	100%

From our results, we can see the BP (44%) and the URP (56%) project team members expended a proportionally larger amount of their meeting time discussing actions related to projected reality while the CP (25%) project team members spent less.

Events of unknowingness triggered the team's conversations about projected reality, which were discussions about how to interpret people's actions and events and proposed actions that should be taken in response to the event. Even though the CP (23%) project team was encountering a somewhat greater number of experiences that required interpretation than the BP (17%) and URP (10%) projects, the BP and URP teams were developing and debating many more interpretations when those events occurred than the CP team did.

We provide the following examples of the BP and the URP projecting reality in which team members are applying motives to individuals without knowing those motives to explain their experience:

In this unexpected event of the URP project, the customer has asked for an onsite demonstration which has flustered the team and they try to explain it among themselves:

I get where you're heading because we've invested so much time and energy into this and we got to a point where we essentially kicked off implementation and we got accounts confirmed back from them and we're ready to move forward. And then all of a sudden you left for paternity leave and they were like oh we're going to pull one over on all of them who are still there and they were like well we can't move forward until xyz is done.
— URP Implementation Manager

In this unexpected event of the BP project, the team feels that they are unable to make progress and begin questioning the organizational structure of the project team:

I think what the problem is, is there's so many TD and O parts and we need one primary, TD, TD and O lead. One person who owns the entire TD and O, who's responsible for the entire, you know, side and make sure that all these moving parts are happening, is making sure these meetings are happening, is making sure the decisions are being made, are documenting it. Like, I just feel like that's where we need somebody, someone has to be accountable there. — BP Business Analyst

Conversely, the CP (30%) team exhibited a proportionally greater number of discussions related to interpretive symmetry in which the team members worked to align their knowledge across functions and with customers and partners while the BP (16%) and URP (13%) teams had fewer discussions about aligning. Similarly, discussions about processes among the CP (14%) team members, whether it was about the existing process or their modifications, was proportionally greater than for the BP (7%) and URP (6%) project teams.

We provide the following as an example of interpretive symmetry within the CP team as the project manager and reporting analyst must develop a new process to deal with an unexpected event which requires they access reporting:

PM: But the other view in salesforce that I can use to look at those real quick.

RA: The open leads?

PM: Yes, the ICE, any Canadian leads that we would have.

RA: Yeah. I can get those. I can just build a report to show what's in there.

PM: Do you mind shooting it over to me or just tell me the name so I can look at it.

RA: No problem.

Analysis of project goals relationship to the organization. In the case of projects, the focal object of reality is the project goal. We had previously set out that for shared reality to exist, actors must have shared beliefs about the project goal, and they must have awareness and a connectedness of these beliefs with each other. In order to explain differences in whether a team had a shared reality or not, we examined the project goal as

it related to organizational purpose. We used the following definition from the firm's website as our reference point for their organizational mission and purpose.

So, who exactly is Payments Co.? We're the ones who believe there's a smarter more efficient way for middle-market businesses to process invoices and make payments; who've created a suite of software and service solutions to fully automate your accounts payable and payment process; and who, as your ally, want to liberate your AP from all that paperwork that's slowing down your business and costing you money.

As we began this first part of our analysis, we sought out dialogue that demonstrated how the teams referred to and understood their project goal. We have previously defined the project goal as their object of reality. We begin first with the CP project which was considered to be successful by the organization. In this excerpt, the outgoing PM has paused a conversation to provide the new PM who is transitioning onto the project with background on the motivation for the CP project. We separately documented in a preliminary conversation with the PM that two of the largest customers of the organization are multi-national firms and therefore have asked that the organization make their solution available in Canada.

You guys heard the money transmitter license or being viewed as a money service business, whatever it's called, MSB....anytime you mention it in a sentence with a Canadian bank, they mainly, it throws up a red flag...so we're trying to build partnerships in Canada ... because our model is a trust fund model. — CP Project Manager

We are provided first with an identity statement about the organization, that they are a money service business and that this identity has created a problem for the organization in providing services to their Canadian clients, which is the project's goal. It follows that this part of this project's purpose is to help build partnerships in Canada so that the organization can provide services in Canada to meet their client's need. Through

this statement, an individual team member can see the connectedness they share with their customer, the organization, their functional team, and the project team by achieving their goal. Their goal restated is to help our customers use our organization's payment services in Canada.

The second project, URP, was starting a third attempt at implementation with a customer following two previous failures at different customer sites. A new project manager has recently been assigned but the remainder of the team shares the history of the past failures. We provide the following statement as a representation of this team's understanding of their goal. This statement was made in response to the Implementations Manager questioning the Product Manager on the reason for the project.

It was a renewal tactic. Their contract was up for renewal and we were trying to figure out what we could lob their way to keep them as our customer. But it was it was kind of a mutual like hey I have a problem you guys provide a solution and you keep me as a customer right. — URP Product Manager

In the case of the URP project, one sees that the goal is to come up with a solution that will make the customer happy, and the motivation for that goal is to keep the organization from losing the customer. In this case, it is not clear to the team exactly what it is they must do that will result in the customer being happy, nor how this, in turn, will allow the organization to keep them as a customer. Therefore, each team member, when attempting to consider this goal from the shoes of the customer or the organization, is provided with little guidance into what that object of reality is nor how they as an individual can relate to it. There is no opportunity for the team to develop a shared belief about the object in the absence of the definition of the object of reality to begin with; there are too many possibilities.

We also note from the company's identity statement that they consider themselves to be an ally of their customers. Given that the motivation for this project was to "save" the customer, we consider that the relationship between the organizations has shifted from one that was built on an alliance to one that is basically become prove your value to me. If team members usually expect to serve in the role of an ally with their customer, this dynamic creates a disconnect beginning with the customer through the organization and down to the employee.

The third project, BP, is starting a second attempt to implement a project that has failed. We provide the following interaction from their first recorded team meeting as an example of their goal statements. In this dialogue, they are referring to a set of four different applications that have been discussed in the earlier part of the meeting.

There's no singular driver for all of them. — BP Technology Manager

We're expanding the scope of this feature into an entire like platform offering and it's... We just need to get this feature into the application.
— BP Product Manager

Prior to this recorded meeting, the PM had stated that the goal of the project was to build an adaptable technology platform. This platform would allow the organization to target larger customers by building solutions more rapidly to meet its demand. It is evident from these three differing statements about what the goal is that there is not a consensus. However, it does appear that there may be a common motivation, which is to gain a new and larger client base. Still, it is not clear what the organization must do to achieve this, which is their missing object of reality. In the absence of an agreed-upon object of reality, the team will not be able to share in their beliefs about that reality. We also refer back to their organizational identity, which stated that they see themselves as

supporting middle-market sized firms. If there is an intended shift from middle-market to larger firms as a target, then this is, in essence, a significant shift from how the firm has seen itself historically.

Measures of identity. Given that a shared reality requires that teams are connected in their beliefs about an object of reality, we would expect that this connectedness over a belief would serve to also strengthen the connectedness the team felt with those around them. We used the affective measure of identity to see if there were differences between teams with a shared reality and teams without a shared reality as they related to their functional team, their project team, and the organization as a whole as well as how the strength of identity shifted across time. We present the results in Table 20 from the first survey taken at the start of the study and Table 21 from the second survey taken at the end of the study.

TABLE 20
Team Affective Identity at Study Beginning

			Identity Measured			
	Participants *	Respondents	Response Rate	Functional	Project	Organizational
BP	12	7	58%	1.3	1.5	1.6
URP	6	5	83%	1.7	1.7	2.0
CP	25	19	76%	1.5	1.9	2.1
Total	31	24	77%	1.5	1.7	1.9

Measurement Scale (1) Strongly Agree <-----> Strongly Disagree (5)

* The survey was sent to the originally identified participants which did not reflect the same participants in the project recordings due to exits and new entries and non-participation.

TABLE 21
Team Affective Identity at Study End

			Identity Measured			
	Response		Rate	Functional	Project	Organizational
	Participants *	Respondents				
BP	12	5	42%	1.1	1.7	1.8
URP	6	5	83%	1.8	2.4	2.3
CP	25	11	44%	1.5	1.9	2.2
Total	31	16	52%	1.4	2.0	2.1

Measurement Scale (1) Strongly Agree <-----> Strongly Disagree (5)

Our expectation at the outset of this study was that functional identity in a project team setting would be stronger than a project team identity, and this supposition was born out by the survey data both at the beginning of the study and at the end of the study for all three project teams. We note that organizational identity was the weakest, and we reason that this makes sense, given that an organization is a less tangible abstract entity. It is important to understand that a team member's first alliance is to their functional team. We found in our second study that functional team identity supported project success. As this identity is weakening, the performance of the team will also be weakening.

Our data from these measures of identity also shows that in the case of the BP and URP project, which had misaligned goals, there was a weakening of all forms of identity. There was one exception for functional identity on the BP team, and we have been able to isolate this to an outlier respondent who rated their functional team very low in the first survey and very high in the second which may be attributable to some unhappy instance at the time of the survey-taking that was later resolved. We contrast these weakening identity results with the stable results for the CP team whose project is aligned with organizational objectives, and that is a performing team.

The movement in these measures of identity serve to provide another point of corroboration about the effects of misaligned goals. Our analysis of the project team meetings indicated that the BP and URP projects were questioning each other's identities, and team members found themselves having to explain their roles to each other and also be challenged on their roles. This leads the entire project team to not only feel disconnected to the goal but also to each other and the organization and affects the overall ability of the team to perform.

The problem we set out to understand was what is the difference between cross-functional project teams that seem able to successfully establish alignment or shared reality while others continue to struggle—often referred to getting on the same page together. Many scholars have looked at this alignment problem from the perspective of interpersonal conflict (Hinds & Mortensen, 2005; Lovelace et al., 2001). Our premise, when engaging in this research, was that we needed to take a look at the contextual issues that, in essence, set the stage for the play as it unfolds. With insights from this study, we hoped that the findings would help organizations become more productive and individuals to suffer less from the frustration experienced when they are not able to progress towards their goal.

Discussion

The problem we set out to understand was what is the difference between cross-functional project teams who seem able to successfully establish alignment or shared reality while others continue to struggle. Often referred to getting on the same page together. Many scholars have looked at this alignment problem from the perspective of interpersonal conflict (Hinds & Mortensen, 2005; Lovelace et al., 2001). Our premise,

when engaging in this research, was that we needed to take a look at the contextual issues that in essence set the stage for the play as it unfolds. With insights from this study, we hoped that the findings would help organizations become more productive and individuals to suffer less from the frustration experienced when they are not able to progress towards their goal.

Our findings help us to explain why the initial setup of the goal and its motivation is so central and important to how the individual actors are able to orient themselves to it. The theoretical foundations of constructed reality provided us with a framework to stand that project goal up, separate it out and consider it as an independent object. In doing so, we could then see the importance of its representation as not only an object of reality for the team members but also as an object that had a relationship within the firm's purpose. For team members to believe in that object of the goal, it must make sense to them in the way that they also understand the firm. This is where misalignment begins and disbelief becomes rooted. Team members must believe in the goal, and they cannot be asked to just take it on faith in order to establish a relationship.

We see that in the event that a team does not establish belief in their goals, they are trapped in a never-ending process of continuous interpretation as each new event unfolds. Interestingly they do not question their own role in relationship to the goal as much as they question the roles and motives of others on the team. Their approach to gaining understanding is akin to pushing the proverbial sand around in the sandbox without coming to any conclusions while their teammates are painted with many different and unpleasant motivations.

In juxtaposition, the project team that has a shared reality and is also experiencing so many unknowns is able to quickly interpret them without ever having to remember what the goal they are pursuing is and why. We liken this to the mental categorization process that Bruner has described in which we categorize an object and develop essential short cuts of interpretation that provide us with an efficient means of getting through our lives (Bruner, 2017). The members of a project team with a shared reality when faced with unexpected events looked inward at their own processes and understandings to try to adapt as needed.

An interesting contrast between the teams was that when the answer to a question was not known team members with a shared reality simply planned to ask the person who did. While in the case of the project teams without shared reality the team members would indulge themselves with speculation about what the answer might be or in the case of a person's motive, what their motives might be.

What these findings imply is that while a project team may be quite busy and certainly has the ability to continually engage in what may seem on the surface to be a productive problem solving dialogue and may even actually produce things, millions of dollars are being wasted on something that is not grounded in a shared reality. Organizations would do well to pay close attention to how goals have been defined in the initial stages of a project to assess how well they represent the true identity of the organization. This is not always an easy thing to do when things like the loss of a customer is at stake.

Limitations and Future Research

One of the limitations of this paper is that the coding was conducted by a single researcher, and the results of the coding were influenced by the coder's subjectivity. An additional limitation of this study is that it was conducted on a single firm, which limits the generalizability of our results. A third limitation is that the study was conducted over a limited time span in the project life cycle such that the research team was not able to see whether goals may have been adapted over time and whether those shifts might achieve the needed alignment to support shared reality.

This study was the first that we are aware of that took shared reality theory and tested its use empirically. Additional studies on the application of shared reality theory are needed to fully develop an understanding of this phenomena. In this study, our data showed and we have articulated the interrelationship between reality and identity, but that was not our focus here. The shared reality perspective could provide identity researchers with the opportunity to consider how identity is affected by the structures of our shared reality and extend our knowledge there.

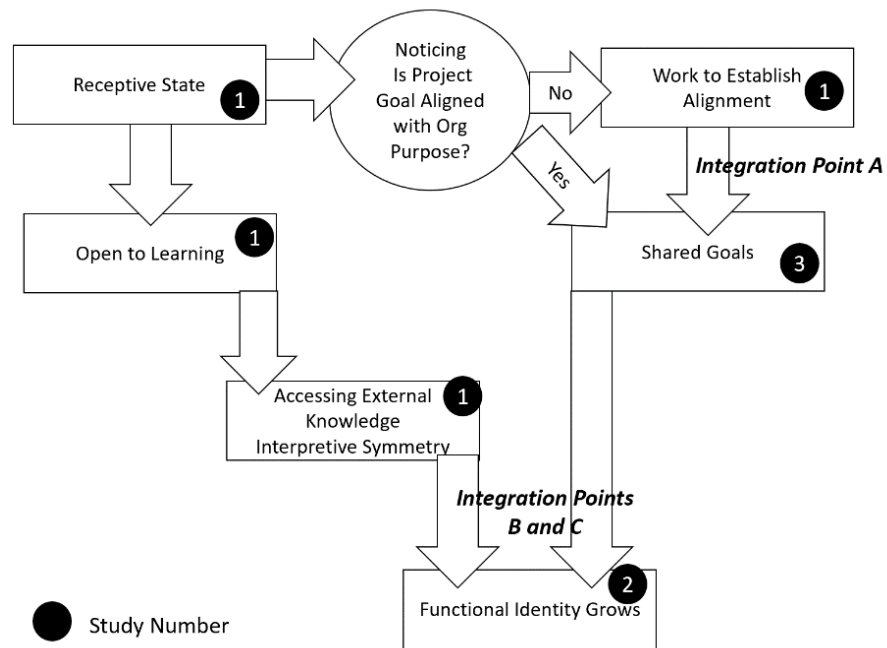
CHAPTER 7: INTEGRATION AND DISCUSSION

Integrated Findings

The theory of shared reality takes the view that we actively construct reality in relationship with those and the environment around us. Embedded within this concept of a shared reality is social identity. The reality we actively construct carries within it our beliefs about our role and the role of the others with whom we are negotiating along with our respective ideas about the organization. This theory provided our research with a viewpoint from which to consider relationships in the context of constructing reality as individuals and teams navigated their knowledge boundaries to achieve a shared reality.

We approached this study with a view to consider the construct of a cross-functional project team from multiple levels of its context: project, functional, and organizational. This multi-level view of our mixed-methods study provided us with an opportunity to connect the dots across levels to consider and validate different levels of connection from a systems design perspective. What follows is a discussion of the results of our mixed-methods study from an integrated perspective. We present the integration points in Figure 9.

FIGURE 9
Integrated Findings



Integration point A: Project to organization – receptiveness and goal alignment.

In Study 1, we found that a heightened awareness and acceptance that the project team did not “know” and needed to learn which we termed receptiveness supported the team’s ability to achieve successful outcomes. At this point of integration for this study, we also see that in this receptive state found in Study 1, project team members are empowered to recognize and challenge goals that are not aligned within the organizational context found in Study 3. When considering the findings from these two studies together, we can see both the necessity of a project team that is intently aware of its environment so that it can actively notice and consider the project goals alignment with how they understand the organization. We provide the following excerpt from Study 1 that expresses this moment

of awareness that there is a misalignment between the goal and the organizational purpose and the need to change it.

I don't think it was a specific decision that I made, but I had a meeting actually with my boss. I started to say, 'We keep hearing these things. It's not going to work for them. We don't have the ability to implement these things.' We actually had a conversation around how unreceptive to feedback, as an organization we are, and that that mentality needs to start to change. It may have been an influence from her, even just saying it, 'We need to change how we think about it.' Then that kind of flipped the switch for me of, 'Yup, you're right. That totally makes sense.' It wasn't necessarily an action item, it just started to be a mindset of being more receptive to that feedback. — Payments Operations Manager

Integration point B: Project to function – interpretive symmetry and functional identity. In our first study, we developed a model of interpretive symmetry to describe the way in which functional teams move back and forth between functional team and project team to adapt and align their routines and processes in support of project goals. This requires the functional team members to actively verbalize or represent through boundary objects their normative practices so that they can be adapted. We consider this finding with the results of our second study which showed that increasing functional identity increases performance when mediated through learning. Our Study 1 finding helps us to explain how this verbalization of routines by the functional team's surfaces and strengthens their own identity as they are learning about each other's roles. We know that learning increases team performance (Edmondson, 1999). From this study, we now understand what type of learning is benefiting cross-functional project teams. We present the following excerpt from an interview that discusses the verbalization of norms and practices in a successful project from Study 1.

So, the first day, it was really talking about what we'd done, how it improved things, how we have the capacity, and our levels of service that we give the

business. The second day was then talking to us, telling us what they do and how they do it differently. And then we then broke out in, on the third day, when we took what they did, marry it to what we did, and then the fourth day was us showing a combined group together on how we would work together to do it. And some of it was putting together the charts of all the different tasks that needed to be done, and some of them would stay and some of them would be moved to the UK. So, it was a cut up, and so at the end of the fourth day, we had a game plan of agreement about what are target state looked like. And everybody in the room, could actually tick and tie everything back. And you could look in the eyes of someone and believe that the ability to be able to service that. — Back Office Operations Manager

Integration point C: Function to organization – Functional identity and shared

goals. In the second study, we found that a strong functional identity positively affects successful outcomes when mediated by learning from another discipline. At the same time, in Study 3, we found the reality of the project goal must accurately represent the reality of the organization. The goal is a representation of the organization as the team members understand it from their functional perspective. Without this alignment, team members are unable to relate to the goal nor to each other. In the absence of alignment, identity is called into question because of the relational nature of functional identity and role to organizational context. This prevents the team from making meaningful progress on the project and is consistent with the opposite findings of Study 2 which found that identity positively affects successful outcomes.

Study 3 also provided a data point with the identity survey demonstrating that in the event of an unaligned project that will likely be unsuccessful, team member identity is weakening across all measures of identity. We provide the following dialogue from Study 3 that exemplifies how identity comes into question when a project team does not recognize and address the fact that they do not have goal alignment.

I totally agree and I don't know who the right person is to drive that because overall, what you're referring to, is project management, right? It's about driving to completion. It... I know you're shaking your head but I think it is. Somebody needs to hold people accountable and say "I need you to be able to sign off. Is it ready to be signed off?" — Development Manager

CHAPTER 8: CONCLUSION

Research Contributions

This research contributes to our knowledge on shared reality, social identity, and their integration through the process of social construction by using theory to conduct a series of empirical studies on cross-functional team communication. Our research demonstrated that the construction of a shared reality determined whether a project would be successful or not. Social construction provides us with a way to conceptualize the interactions of project teams that are grappling with social boundaries of identity that are represented within the team's unique norms and practices.

With these considerations in mind, our study's findings contribute to social construction theory through our development of a process model of interpretive symmetry. This model builds on the ideas of social construction by demonstrating the interactions through which functional team members are facilitating the construction of a shared reality within their own disciplinary team and the project team simultaneously. The ability of a cross-functional team to elicit and reconcile representations of reality is a key capability of the project team and is central to their successful performance.

The initial scope of our analysis and our first study considered the dual accountabilities of the functional team on a project team. The project team participants having membership both with their functional team and their project team requires they act as a boundary spanner moving their functional knowledge, norms, and routines back and forth between project and function to evolve them. Our process model of interpretive symmetry thus contributes to the literature on cross-boundary teaming by presenting the

dynamics of an individual's identification with a group and the critical link to their functional team in an intra-team communication context.

In this first study, we also identified that receptivity is an important initial starting state for cross-functional project teams to engage in learning between functions and from external sources. We defined a receptive state as the awareness and acceptance that the team does not know and must learn. We wish to differentiate this receptive state from the term mindfulness, which is defined as being consciously aware and focusing on the present moment (Good et al., 2015). Conscious awareness implies deliberate action to achieve a specific state to notice the environment. While in our study, we found that being in a receptive state does not require a conscious effort because it has been internalized as a knowing, as an individual's receiver is set to "On" subconsciously.

Our finding on receptive attention contributes to the literature on cross-boundary teaming by identifying an important starting state for project teams to be able to access external knowledge, whether it is from a team member from a different function or knowledge that is external to the project team or firm. It also explains how teams become aware of a project goal's misalignment with firm purpose and thus contributes to research on shared goals and shared reality.

Our second study furthered our knowledge on the role of functional identity and intra-team communication by demonstrating that increasing functional identity increases our ability to understand someone from a different discipline. This study provided another data point for the debate on whether diversity is beneficial or hinders team performance. By revealing the relationship between functional identity and the ability to learn from another discipline as a mediator, we were able to show under what conditions

diversity in the form of functional expertise or identity is supporting successful outcomes. This study contributes to our knowledge on diversity as well as to the knowledge on cross-boundary teaming.

This same finding about the relationship between functional identity and performance has also provided us with insight into what type of knowledge or learning is necessary for teams to achieve performance. This sharing of norms and routine practices across functional boundaries supports cross-functional team coordination by helping the project team to understand how they relate to one another in the current and evolving organizational system they are engaging in.

In our final study, we revealed the circumstances under which dialogues about functional identity are supported and shared and when they are not. Understanding that this communication is important in the process of reality construction, we identified a specific factor that inhibited this dialogue—the project goal. Shared goals which have received little attention in the literature have been shown to be a factor in successful outcomes for teams (Carson et al., 2007). With this study, we were able to observe how it was essential that project goals be aligned with firm purpose in order for the project team to construct a shared reality. This study has helped us to understand the connection between a goal and functional identity in an intra-team context.

Organizations are relational systems where individuals understand their role and their actions within this relational context. Goals also have a relation to this system, and when they are not aligned, team members are unable to understand how they relate to the goal within an organizational context. This study contributes to the literature on shared goals demonstrating a necessary condition for a team to develop a way of knowing a

shared goal in common. This study also contributes to the literature on cross-boundary teaming by highlighting another important starting state and a necessary condition for achievement in order for teams to achieve success. This study also contributes to the literature on shared reality by providing empirical evidence of how shared reality is created in the form of goals within an organizational context.

Limitations

Our first study was limited by using the interviewee's perception of success, which is a subjective measure. Past studies have shown that an individual may perceive a project as successful when the team is collaborating well, yet this does not always mean that the project was considered successful from outside perspectives (Ancona & Caldwell, 1992a). We addressed this limitation through additional studies that corroborate our initial finding about the process model of interpretive symmetry and the role of cross-functional team communication, functional identity, and learning from other disciplines as important factors in achieving team performance.

Our third study was limited by the sample size with only three teams in a single company in the study, which does not allow us to generalize our results. However, this study was meant to corroborate and further elucidate the findings from Studies 1 and 2 such that the findings about functional identity further enrich our understanding of how and when functional identity is elicited. This study also provided us with initial evidence on the role project goals play in the creation of reality.

Our third study was also limited because the coding was performed by a single researcher, which introduces subjectivity into the interpretation of the data. Again, because this study was used to corroborate other findings, its strength comes through the

ability to confirm original findings from other studies and provide additional explanatory power to those studies.

Implications

We considered the failure rate of projects, particularly those with technology as were studied here and the growing utilization of the cross-functional project team form as an important reason to pursue this line of research. Given the multiple levels studied and the many findings this study surfaced, we argue that the most important point of this study for a firm to consider is whether project goals are aligned with firm purpose. This is the starting point from which shared reality can or cannot be created among cross-functional team members. From a practice perspective, this becomes an issue for organizational leadership and a firm's ability to govern the project work the firm is employed in.

Equally important to consider is the effect that unaligned goals have on a team's sense of functional identity and their struggle to understand their own and their teammates' roles in the absence of a goal grounded in a shared reality. When we consider the lack of engagement of employees in today's workforce, we might consider that this very lack of shared reality plays an important part in their feelings of disconnectedness with each other and the firm (Harter, 2018).

Based on our findings, we recommend that firms struggling with poor project performance can improve their cross-functional project teams' results by building organization-wide communication capabilities. We propose conceptually the implementation of a Shared Reality System™ which would encompass a multi-level strategy to build firm communication capability. This system would be customized to the

context of the unique reality of each individual firm and implemented in multiple phases. The first phase of the Shared Reality System™ would focus at the senior management level within the firm to establish a dynamic process for articulating and integrating firm purpose.

In this phase, a framework would be created and supporting processes defined that provide a feedback loop between organizational stakeholders and senior leaders responsible for ensuring the firm's vision is being continuously articulated, considered, validated, and integrated throughout the organization. This process would be unlike the typical vision and values processes that result in a static document that is completed, distributed to employees and shelved. The system is meant to be a series of actions and processes that are dynamic and occur in response to a conscious understanding of the stated purpose, strategies, and intentions of the firm.

The second and third phases of this system would focus on employee training, specifically the functional team members who are engaged in project delivery. The SRS Boundary Spanner Training Package™ would utilize a train the trainer model and be customized based on an assessment of the firm's current communication capabilities. Training focuses on the roles and responsibilities defined within the organizational context with integration into systems and business processes that will be created or modified to sustain role enactment.

With the implementation of a Shared Reality System™, we believe a firm would develop the needed level of organizational communication capabilities to ensure the successful outcome of their cross-functional project teams.

Future Research

There are few studies that have availed themselves of the construct and ideas of shared reality, but the perspective of this theory provides us with a very insightful approach to analyzing interrelationships. We believe that shared reality can provide valuable insights into other aspects of management research, particularly as it might be applied to ideas about leadership.

When we consider the concept of a shared goal and its alignment with a firm's purpose, we are left with the unanswered question of how do unaligned goals come into being? What are the factors that affect a goal's formation and what processes do different organizations use to both create and govern them? Who are the appropriate stakeholders in the creation of the organization's purpose and their goals?

We believe much can be learned from this new line of inquiry not only for focusing the firm on productive efforts but for the emotional well-being of its employees. We consider even in today's business world organizations that are acting in ways that are inconsistent with their statements of vision and values, as we have seen in the recent financial crisis. We understand that goal alignment not only affects those inside the firm but equally those outside the firm.

APPENDIX A

Study 1 Coding Summary

Step I - Open Coding				Step II - Main Themes	Step III - Main Theories	Theory Summary
Common Vision - Expectations (108)				Creating the Vision (21) Agreement with the vision (16)	Shared Reality	Shared Reality Boundary Spanning Boundary Objects Mindfulness
Challenging the vision (5)	Evolving Vision (7)	Organized to support the vision (10)	Communicating the vision (7)			
Conflicting visions (6)	Expectations of the vision (14)	Agreement with the vision (16)	Importance of the mission (5)			
Creating the vision (21)	Impacting the vision (1)	Commitment to the vision (12)	Working without a vision (4)			
Communication (207)				Method of Communication (26) Not Communicating (16)	Boundary Objects Shared Reality	
Ability to persuade (9)	Negative communications (9)	Escalation (9)	Indirect communication (5)			
Common grounding (1)	Back and forth (6)	Forceful (3)	Method of communication (26)			
Common language (7)	Discussing actions being taken (7)	Relaying experiences (6)	Non-verbal (3)			
Correcting (4)	Discussing concepts (5)	Repetitive (3)	Tools (5)			
Hearing (1)	Downward (3)	Too much communication (7)	Lacking (5)			
Interpreting what is being said (3)	Positive communications (1)	Upward (3)	Not communicating (16)			
Listening (8)	Quality of the conversation (8)	Validation (1)	Not hearing (5)			
Hierarchical (8)	Setting expectations (6)	Environments and communicating (8)	Restricted (6)			
		Effectiveness (6)	Stopped discussing a topic (4)			
Knowledge - Existing (Base) (74)				Gathering knowledge together to share (12) Having Experience (23)	Boundary Spanning Shared Reality	
Conflicting data (2)	Ease of knowing (4)	Having Experience (23)	Knowledge in picture form (1)			
Consciousness of understanding (2)	Fully using knowledge (3)	Knowing how to work together (1)	Knowledge of constraints (1)			
Credibility of knowledge (9)	Gathering knowledge together (12)	Knowing the outcome (8)	Bigger picture (1)			
			Who has the knowledge (7)			
Knowledge - Creation (Solutions)				Considerations or constraints (20) Environment (22) Ideas (27) Methods for creating knowledge (32) People involved (20) Finding a Solution (42)	Shared Reality Boundary Objects Boundary Spanning	
Applying lessons learned (9)	Evolution of ideas (8)	People involved (20)	Starting from the known (3)			
Building (4)	Ideas (27)	Probing (5)	Alternate scenarios (1)			
Capturing the ideas (8)	Ideas designed for temporary use (7)	Process (19)	Time needed (5)			
Choosing a starting point (9)	Implementing something new (2)	Questions that are asked (13)	Uncertainty (2)			
Co-creation (8)	Iterations (10)	Resisting (2)	Contradicting (5)			
Considerations or constraints (7)	Methods for creating knowledge (32)	Results of testing the solution (13)	Environment (22)			
	New understanding (13)	Finding a Solution (42)	Noticing (4)			
Knowledge - Learning (New) (93)				Acknowledging the Need to Learn (12) Comprehension (25) Investigation (13)	Shared Reality Mindfulness	
Know need to learn (12)	Consolidating learnings (1)	Infering from the data (1)	Learning while doing (8)			
Comprehension (25)	Discovery or Findings (8)	Inquisitiveness (5)	Memory (12)			
Confirming Understanding (5)	Incorrect Information (1)	Investigation (13)	Researching (3)			
Knowledge or Ability - Lacking				Comprehension - Lacking (22) Lacking Experience (17) Missing Information (25)	Shared Reality Boundary Spanning	
Ability to problem solve (5)	Don't know where to start (1)	Lacking Experience (17)	No time to learn (1)			
Comprehension - Lacking (22)	Negative Effects of (8)	Language barriers (5)	Suppressing knowledge (2)			
No Context (3)	Hidden information (5)	Lost the knowledge (6)	Trial and Error (1)			
Depth of understanding (9)	Incorrect assumptions (4)	Missing Information (25)	Unaware (2)			
Doesn't know how it works (6)	Incorrect Information (13)	Never done before (8)	When to expect something (1)			
			Where to find information (3)			
Knowledge Sharing (130)				Common Understanding (17) Process of Knowledge Sharing (41)	Shared Reality Boundary Spanning	
Ability to knowledge share (7)	Common Understanding (17)	Conditions for knowledge sharing (6)	Process (41)			
Are knowledge sharing (2)	Comprehending - Not (6)	Experience of knowledge sharing (13)	Providing context (2)			
Collaboration (12)	Interpreting meaning (13)	Not Knowledge Sharing (11)				
Working together (579)				Accountability (86) Collaboration (89) Relationships (64)	Shared Reality Boundary spanning	
Accountability (6)	Collaborative - not (37)	Initiative (11)	Relationships (64)			
Accountability (86)	Competence (19)	Initiative - Lacking (2)	Resistance (25)			
Agreement (7)	Controlling and Directing (26)	Leadership (40)	Resource limitations (2)			
Anticipation (3)	Disagreement (11)	Meeting Needs (16)	Separateness (6)			
Assumptions (5)	Engagement - Not (3)	Network creation (8)	Team changes (19)			
Autonomy (9)	Environment (28)	Non-conflictative (1)	Trust (23)			
Collaboration (52)	Help (19)	Overconfident (1)	Uncertainty (5)			

REFERENCES

- Albert, S., Ashforth, B. E., & Dutton, J. E. 2000. Organizational identity and identification: Charting new waters and building new bridges. *Academy of management review*, 25(1): 13-17.
- Allen, N. J., & Meyer, J. P. 1990. The measurement and antecedents of affective, continuance and normative commitment to the organization. *Journal of occupational psychology*, 63(1): 1-18.
- Alvesson, M., Lee Ashcraft, K., & Thomas, R. 2008. Identity matters: Reflections on the construction of identity scholarship in organization studies. *Organization*, 15(1): 5-28.
- Anbari, F. T., Bredilett, C. N., & Turner, J. R. 2008. Perspectives on research in project management, *Academy of Management Proceedings*, vol. 2008: 1-6: Academy of Management.
- Ancona, D. G. 1990. Outward bound: Strategic for team survival in an organization. *Academy of Management Journal*, 33(2): 334-365.
- Ancona, D. G., & Caldwell, D. F. 1992a. Bridging the boundary: External activity and performance in organizational teams. *Administrative Science Quarterly*, 37(4): 634-665.
- Ancona, D. G., & Caldwell, D. F. 1992b. Demography and design: Predictors of new product team performance. *Organization Science*, 3(3): 321-341.
- Argote, L., & Ingram, P. 2000. Knowledge transfer: A basis for competitive advantage in firms. *Organizational Behavior and Human Decision Processes*, 82(1): 150-169.
- Ashforth, B. E., & Mael, F. 1989. Social identity theory and the organization. *Academy of Management Review*, 14(1): 20-39.
- Ashforth, B. E., Rogers, K. M., & Corley, K. G. 2011. Identity in organizations: Exploring cross-level dynamics. *Organization Science*, 22(5): 1144-1156.
- Astley, W. G., & Zammuto, R. F. 1992. Organization science, managers, and language games. *Organization science*, 3(4): 443-460.
- Bar-Tal, D. 2000. *Shared beliefs in a society: Social psychological analysis*: Sage Publications.
- Baron, R. S., Hoppe, S. I., Kao, C. F., Brunzman, B., Linneweh, B., & Rogers, D. 1996. Social corroboration and opinion extremity. *Journal of Experimental Social Psychology*, 32(6): 537-560.

- Bechky, B. A. 2003. Sharing meaning across occupational communities: The transformation of understanding on a production floor. *Organization Science*, 14(3): 312-330.
- Bechky, B. A. 2006. Gaffers, gofers, and grips: Role-based coordination in temporary organizations. *Organization science*, 17(1): 3-21.
- Bechky, B. A., & Chung, D. E. 2018. Latitude or latent control? How occupational embeddedness and control shape emergent coordination. *Administrative Science Quarterly*, 63(3): 607-636.
- Bell, S. T., Villado, A. J., Lukasik, M. A., Belau, L., & Briggs, A. L. 2011. Getting specific about demographic diversity variable and team performance relationships: A meta-analysis. *Journal of Management*, 37(3): 709-743.
- Berger, P., & Kellner, H. 1964. Marriage and the construction of reality: An exercise in the microsociology of knowledge. *Diogenes*, 12(46): 1-24.
- Berger, P. L., & Luckmann, T. 1991. *The social construction of reality: A treatise in the sociology of knowledge*: Penguin UK.
- Blumer, H. 1986. *Symbolic interactionism: Perspective and method*: University of California Press.
- Boland Jr, R. J., & Tenkasi, R. V. 1995. Perspective making and perspective taking in communities of knowing. *Organization Science*, 6(4): 350-372.
- Bowker, G., & Star, S. L. 1999. *Sorting things out: Classification and its consequences*. Cambridge, MA; London: The MIT Press.
- Brown, A. D. 2006. A narrative approach to collective identities. *Journal of Management Studies*, 43(4): 731-753.
- Brown, A. D., & Lewis, M. A. 2011. Identities, discipline and routines. *Organization Studies*, 32(7): 871-895.
- Brown, A. D. 2015. Identities and identity work in organizations. *International Journal of Management Reviews*, 17(1): 20-40.
- Bruner, J. 1991. The narrative construction of reality. *Critical Inquiry*, 18(1): 1-21.
- Bruner, J. 2017. *A study of thinking*: Routledge.
- Bruner, J. S. 1957. On perceptual readiness. *Psychological Review*, 64(2): 123-152.
- Butler, J. 2005. *Giving an account of oneself*: Oxford University Press.

- Cameron, J. E. 2004. A three-factor model of social identity. *Self and Identity*, 3(3): 239-262.
- Carlile, P. R. 2002. A pragmatic view of knowledge and boundaries: Boundary objects in new product development. *Organization Science*, 13(4): 442-455.
- Carson, J. B., Tesluk, P. E., & Marrone, J. A. 2007. Shared leadership in teams: An investigation of antecedent conditions and performance. *Academy of Management Journal*, 50(5): 1217-1234.
- Charmaz, K. 2014. *Constructing grounded theory*: Sage.
- Chatterjee, S., Hadi, A., & Price, B. 2000. *Regression analysis by example*. New York, NY: Wiley.
- Chesebro, J. L., & McCroskey, J. C. 1998. The development of the teacher clarity short inventory (TCSI) to measure clear teaching in the classroom. *Communication Research Reports*, 15(3): 262-266.
- Chow, W. S., & Chan, L. S. 2008. Social network, social trust and shared goals in organizational knowledge sharing. *Information & Management*, 45(7): 458-465.
- Cooper, D., & Thatcher, S. M. 2010. Identification in organizations: The role of self-concept orientations and identification motives. *Academy of Management Review*, 35(4): 516-538.
- Creswell, J. W., & Plano Clark, V. L. 2017. *Designing and conducting mixed methods research*: Sage publications.
- DeChurch, L. A., & Marks, M. A. 2006. Leadership in multiteam systems. *Journal of Applied Psychology*, 91(2): 311.
- Denison, D. R., Hart, S. L., & Kahn, J. A. 1996. From chimneys to cross-functional teams: Developing and validating a diagnostic model. *Academy of Management Journal*, 39(4): 1005-1023.
- Dionysiou, D. D., & Tsoukas, H. 2013. Understanding the (re) creation of routines from within: A symbolic interactionist perspective. *Academy of Management Review*, 38(2): 181-205.
- Echterhoff, G., Higgins, E. T., & Levine, J. M. 2009. Shared reality: Experiencing commonality with others' inner states about the world. *Perspectives on Psychological Science*, 4(5): 496-521.
- Edmondson, A. 1999. Psychological safety and learning behavior in work teams. *Administrative Science Quarterly*, 44(2): 350-383.

- Edmondson, A. C. 2004. Learning from failure in health care: Frequent opportunities, pervasive barriers. *BMJ Quality & Safety*, 13(suppl 2): ii3-ii9.
- Edmondson, A. C., & Harvey, J.-F. 2018. Cross-boundary teaming for innovation: Integrating research on teams and knowledge in organizations. *Human Resource Management Review*, 28(4): 347-360.
- Ellemers, N., Van Rijswijk, W., Roefs, M., & Simons, C. 1997. Bias in intergroup perceptions: Balancing group identity with social reality. *Personality and Social Psychology Bulletin*, 23(2): 186-198.
- Ellemers, N., Kortekaas, P., & Ouwerkerk, J. W. 1999. Self-categorisation, commitment to the group and group self-esteem as related but distinct aspects of social identity. *European Journal of Social Psychology*, 29(2-3): 371-389.
- Ellemers, N., De Gilder, D., & Haslam, S. A. 2004. Motivating individuals and groups at work: A social identity perspective on leadership and group performance. *Academy of Management Review*, 29(3): 459-478.
- Ellemers, N. 2019. Identification and cooperation, *Identification and Cooperation*: Vimeo. <https://vimeo.com/328102748>.
- Faraj, S., & Xiao, Y. 2006. Coordination in fast-response organizations. *Management Science*, 52(8): 1155-1169.
- Feldman, M. S., & Rafaeli, A. 2002. Organizational routines as sources of connections and understandings. *Journal of Management Studies*, 39(3): 309-331.
- Gal, U., Yoo, Y., & Boland, J., Richard J. 2005. *The dynamics of boundary objects, social infrastructures and social identities*: European Conference on Information Systems (ECIS) 2005 Proceedings, 57. May 26-28, Regensburg, Germany. <http://www.ecis2005.de/>.
- Gerpott, F. H., Lehmann-Willenbrock, N., Voelpel, S. C., & van Vugt, M. 2019. It's not just what is said, but when it's said: A temporal account of verbal behaviors and emergent leadership in self-managed teams. *Academy of Management Journal*, 62(3): 717-738.
- Gladstein, D. L. 1984. Groups in context: A model of task group effectiveness. *Administrative science quarterly*: 499-517.
- Glaser, B. G., & Strauss, A. L. 2012. *The discovery of grounded theory: Strategies for qualitative research*. New Brunswick: Aldine Transaction.
- Goffman, E. 2008. *Behavior in public places*: Simon and Schuster.

- Good, D. J., Lyddy, C. J., Glomb, T. M., Bono, J. E., Brown, K. W., Duffy, M. K., Baer, R. A., Brewer, J. A., & Lazar, S. W. 2015. Contemplating mindfulness at work: An integrative review. *Journal of Management*, 42(1): 114-142.
- Hardy, C., Lawrence, T. B., & Grant, D. 2005. Discourse and collaboration: The role of conversations and collective identity. *Academy of Management Review*, 30(1): 58-77.
- Harrison, D. A., Mohammed, S., McGrath, J. E., Florey, A. T., & Vanderstoep, S. W. 2003. Time matters in team performance: Effects of member familiarity, entrainment, and task discontinuity on speed and quality. *Personnel Psychology*, 56(3): 633-669.
- Harter, J. 2018. Employee engagement on the rise in the U.S. *Gallup*.
<https://news.gallup.com/poll/241649/employee-engagement-rise.aspx>.
- Hayes, A. F., & Preacher, K. J. 2013. Conditional process modeling: Using structural equation modeling to examine contingent causal processes. In G. R. Hancock & R. O. Mueller (Eds.), *Quantitative methods in education and the behavioral sciences: Issues, research, and teaching. Structural equation modeling: A second course*: 219-266: IAP Information Age Publishing.
- Higgins, E. T. 1992. Achieving 'shared reality' in the communication game: A social action that create; meaning. *Journal of Language and Social Psychology*, 11(3): 107-131.
- Higgins, E. T., & Pittman, T. S. 2008. Motives of the human animal: Comprehending, managing, and sharing inner states. *Annual Review of Psychology*, 59.
- Hinds, P. J., & Mortensen, M. 2005. Understanding conflict in geographically distributed teams: The moderating effects of shared identity, shared context, and spontaneous communication. *Organization Science*, 16(3): 290-307.
- Hoegl, M., Weinkauff, K., & Gemuenden, H. G. 2004. Interteam coordination, project commitment, and teamwork in multiteam R&D projects: A longitudinal study. *Organization Science*, 15(1): 38-55.
- Hogg, M. A., & Reid, S. A. 2006. Social identity, self-categorization, and the communication of group norms. *Communication Theory*, 16(1): 7-30.
- Hollenbeck, J. R., Beersma, B., & Schouten, M. E. 2012. Beyond team types and taxonomies: A dimensional scaling conceptualization for team description. *Academy of Management Review*, 37(1): 82-106.
- Hu, L. t., & Bentler, P. M. 1999. Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives. *Structural Equation Modeling: A Multidisciplinary Journal*, 6(1): 1-55.

- Ibarra, H. 1999. Provisional selves: Experimenting with image and identity in professional adaptation. *Administrative Science Quarterly*, 44(4): 764-791.
- Ibarra, H., & Barbulescu, R. 2010. Identity as narrative: Prevalence, effectiveness, and consequences of narrative identity work in macro work role transitions. *Academy of Management Review*, 35(1): 135-154.
- Jehn, K. A., Northcraft, G. B., & Neale, M. A. 1999. Why differences make a difference: A field study of diversity, conflict and performance in workgroups. *Administrative Science Quarterly*, 44(4): 741-763.
- Johnson, S. L., Boland, R. J., Cola, P., Aron, D., & Wang, Y. 2018. How cross-functional project teams construct a shared reality to align organizational knowledge. Cleveland, OH: Unpublished work. Case Western Reserve University.
- Kacmar, K. M., Witt, L., Zivnuska, S., & Gully, S. M. 2003. The interactive effect of leader-member exchange and communication frequency on performance ratings. *Journal of Applied Psychology*, 88(4): 764.
- Kane, A. A. 2010. Unlocking knowledge transfer potential: Knowledge demonstrability and superordinate social identity. *Organization Science*, 21(3): 643-660.
- Kellogg, K. C., Orlikowski, W. J., & Yates, J. 2006. Life in the trading zone: Structuring coordination across boundaries in postbureaucratic organizations. *Organization Science*, 17(1): 22-44.
- Knight, D., Durham, C. C., & Locke, E. A. 2001. The relationship of team goals, incentives, and efficacy to strategic risk, tactical implementation, and performance. *Academy of Management Journal*, 44(2): 326-338.
- Kock, N., & Lynn, G. 2012. Lateral collinearity and misleading results in variance-based SEM: An illustration and recommendations. *Journal of the Association for Information Systems*, 13(7): 546-580.
- Kozlowski, S. W., & Bell, B. S. 2003. Work groups and teams in organizations. In W. C. Borman, D. R. Ilgen, & R. J. Klimoski (Eds.), *Handbook of psychology*, vol. 12: 333-375: John Wiley & Sons, Inc.
- Kreiner, G. E., Hollensbe, E. C., & Sheep, M. L. 2006. On the edge of identity: Boundary dynamics at the interface of individual and organizational identities. *Human Relations*, 59(10): 1315-1341.
- Krigsman, M. 2012. Worldwide cost of IT failure (revisited): \$3 trillion.
<https://www.zdnet.com/article/worldwide-cost-of-it-failure-revisited-3-trillion/>.

- Levine, J. M., Higgins, E. T., & Choi, H.-S. 2000. Development of strategic norms in groups. *Organizational Behavior and Human Decision Processes*, 82(1): 88-101.
- Lewin, K. 1948. *Resolving social conflicts; selected papers on group dynamics*. New York, NY: Harper & Brothers.
- Lewis, K. 2003. Measuring transactive memory systems in the field: Scale development and validation. *Journal of Applied Psychology*, 88(4): 587-603.
- Lovelace, K., Shapiro, D. L., & Weingart, L. R. 2001. Maximizing cross-functional new product teams' innovativeness and constraint adherence: A conflict communications perspective. *Academy of Management Journal*, 44(4): 779-793.
- MacKenzie, S. B., Podsakoff, P. M., & Podsakoff, N. P. 2011. Construct measurement and validation procedures in MIS and behavioral research: Integrating new and existing techniques. *MIS Quarterly*, 35(2): 293-334.
- Malone, T. W., & Crowston, K. 1994. The interdisciplinary study of coordination. *ACM Computing Surveys (CSUR)*, 26(1): 87-119.
- Marrone, J. A. 2010. Team boundary spanning: A multilevel review of past research and proposals for the future. *Journal of Management*, 36(4): 911-940.
- Mastrogiacomo, S., Missonier, S., & Bonazzi, R. 2014. Talk before it's too late: reconsidering the role of conversation in information systems project management. *Journal of Management Information Systems*, 31(1): 47-78.
- Matsunaga, M. 2010. How to factor-analyze your data right: Do's, don'ts, and how-to's. *International Journal of Psychological Research*, 3(1): 97-110.
- McDonough III, E. F. 2000. Investigation of factors contributing to the success of cross-functional teams. *Journal of Product Innovation Management*, 17(3): 221-235.
- McGrath, J. E., Arrow, H., & Berdahl, J. L. 2000. The study of groups: Past, present, and future. *Personality and Social Psychology Review*, 4(1): 95-105.
- Merriam-Webster. n.d. goal. <https://www.merriam-webster.com/dictionary/goal>. February 26, 2020.
- Mitchell, R., Boyle, B., Parker, V., Giles, M., Chiang, V., & Joyce, P. 2015. Managing inclusiveness and diversity in teams: How leader inclusiveness affects performance through status and team identity. *Human Resource Management*, 54(2): 217-239.

- Mohammed, S., Klimoski, R., & Rentsch, J. R. 2000. The measurement of team mental models: We have no shared schema. *Organizational Research Methods*, 3(2): 123-165.
- Nag, R., Corley, K. G., & Gioia, D. A. 2007. The intersection of organizational identity, knowledge, and practice: Attempting strategic change via knowledge grafting. *Academy of Management Journal*, 50(4): 821-847.
- Nonaka, I. 1994. A dynamic theory of organizational knowledge creation. *Organization Science*, 5(1): 14-37.
- O'Mahony, S., & Bechky, B. A. 2008. Boundary organizations: Enabling collaboration among unexpected allies. *Administrative Science Quarterly*, 53(3): 422-459.
- Petriglieri, J. L. 2011. Under threat: Responses to and the consequences of threats to individuals' identities. *Academy of Management Review*, 36(4): 641-662.
- Pinto, M. B., & Pinto, J. K. 1990. Project team communication and cross-functional cooperation in new program development. *Journal of Product Innovation Management*, 7(3): 200-212.
- Pinto, M. B., Pinto, J. K., & Prescott, J. E. 1993. Antecedents and consequences of project team cross-functional cooperation. *Management Science*, 39(10): 1281-1297.
- Project Management Institute. 2013. *The high cost of low performance: The essential role of communications*. Newtown Square, PA: PMI's Pulse of the Profession, May 2013. <https://www.pmi.org/-/media/pmi/documents/public/pdf/learning/thought-leadership/pulse/the-essential-role-of-communications.pdf>.
- Project Management Institute. 2017. *Success rates rise: Transforming the high cost of low performance*: PMI's Pulse of the Profession 2017. <https://www.pmi.org/learning/thought-leadership/pulse/pulse-of-the-profession-2017>.
- Project Management Institute. 2018. *Success in disruptive times: Expanding the value delivery landscape to address the high cost of low performance*: PMI's Pulse of the Profession 2018. <https://www.pmi.org/-/media/pmi/documents/public/pdf/learning/thought-leadership/pulse/pulse-of-the-profession-2018.pdf>.
- Richter, A. W., West, M. A., Van Dick, R., & Dawson, J. F. 2006. Boundary spanners' identification, intergroup contact, and effective intergroup relations. *Academy of Management Journal*, 49(6): 1252-1269.

- Riek, B. M., Mania, E. W., & Gaertner, S. L. 2006. Intergroup threat and outgroup attitudes: A meta-analytic review. *Personality and Social Psychology Review*, 10(4): 336-353.
- Riketta, M., & Van Dick, R. 2005. Foci of attachment in organizations: A meta-analytic comparison of the strength and correlates of workgroup versus organizational identification and commitment. *Journal of Vocational Behavior*, 67(3): 490-510.
- Rose, K. H. 2013. A guide to the Project Management Body of Knowledge (PMBOK® Guide)—5th edition. *Project Management Journal*, 44(3): e1-e1.
- Schotter, A. P., Mudambi, R., Doz, Y. L., & Gaur, A. 2017. Boundary spanning in global organizations. *Journal of Management Studies*, 54(4): 403-421.
- Scott, C. R. 2007. Communication and social identity theory: Existing and potential connections in organizational identification research. *Communication Studies*, 58(2): 123-138.
- Seidel, V. P., & O'Mahony, S. 2014. Managing the repertoire: Stories, metaphors, prototypes, and concept coherence in product innovation. *Organization Science*, 25(3): 691-712.
- Sethi, R. 2000. Superordinate identity in cross-functional product development teams: Its antecedents and effect on new product performance. *Journal of the Academy of Marketing Science*, 28(3): 330-344.
- Sherif, M., & Sherif, C. W. 1969. Ingroup and intergroup relations: Experimental analysis. *Social Psychology*: 221-266.
- Simon, H. A. 1996. *The sciences of the artificial* (3rd ed.). Cambridge, MA: MIT Press.
- Söderlund, J. 2011. Pluralism in project management: Navigating the crossroads of specialization and fragmentation. *International Journal of Management Reviews*, 13(2): 153-176.
- Star, S. L., & Griesemer, J. R. 1989. Institutional ecology, translations, and boundary objects: Amateurs and professionals in Berkeley's Museum of Vertebrate Zoology, 1907-39. *Social Studies of Science*, 19(3): 387-420.
- Sveningsson, S., & Alvesson, M. 2003. Managing managerial identities: Organizational fragmentation, discourse and identity struggle. *Human Relations*, 56(10): 1163-1193.
- Tabachnick, B. G., & Fidell, L. S. 2001. *Using multivariate statistics* (4th ed.). Boston, MA: Allyn and Bacon.

- Tajfel, H. 1974. Social identity and intergroup behaviour. *Information (International Social Science Council)*, 13(2): 65-93.
- Tajfel, H. 1982. Social psychology of intergroup relations. *Annual Review of Psychology*, 33(1): 1-39.
- Tajfel, H., & Turner, J. C. 1986. An integrative theory of intergroup relations. In S. Worchel & W. G. Austin (Eds.), *Psychology of intergroup relations*: 7-24. Chicago, IL: Nelson-Hall.
- Teddlie, C., & Tashakkori, A. 2009. *Foundations of mixed methods research: Integrating quantitative and qualitative approaches in the social and behavioral sciences*. Thousand Oaks, CA: Sage Publications.
- Towers Watson. 2012. 2011 – 2012 Change and Communication ROI Study: <https://www.slideshare.net/TowersWatson/towers-watson-2011-2012-change-and-communication-roi-study>.
- Van Der Vegt, G. S., & Bunderson, J. S. 2005. Learning and performance in multidisciplinary teams: The importance of collective team identification. *Academy of Management Journal*, 48(3): 532-547.
- van Knippenberg, D., van Ginkel, W. P., & Homan, A. C. 2013. Diversity mindsets and the performance of diverse teams. *Organizational Behavior and Human Decision Processes*, 121(2): 183-193.
- Vignoles, V. L., Regalia, C., Manzi, C., Gollledge, J., & Scabini, E. 2006. Beyond self-esteem: Influence of multiple motives on identity construction. *Journal of Personality and Social Psychology*, 90(2): 308-333.
- Vygotsky, L. S. 1962. *Thought and language*. Cambridge, MA: MIT Press.
- Weick, K. E., Sutcliffe, K. M., & Obstfeld, D. 2005. Organizing and the process of sensemaking. *Organization Science*, 16(4): 409-421.
- West, S. G., Finch, J. F., & Curran, P. J. 1995. Structural equation models with nonnormal variables: Problems and remedies. In R. H. Hoyle (Ed.), *Structural equation modeling: Concepts, issues, and applications*: 56-75. Thousand Oaks, CA: Sage Publications, Inc.
- Whittington, R., Pettigrew, A., Peck, S., Fenton, E., & Conyon, M. 1999. Change and complementarities in the new competitive landscape: A European panel study, 1992–1996. *Organization Science*, 10(5): 583-600.
- Wittgenstein, L. 1974. *Philosophical Investigations*: MacMillan (First Edition: 1953).

Ybema, S., Keenoy, T., Oswick, C., Beverungen, A., Ellis, N., & Sabelis, I. 2009.
Articulating identities. *Human Relations*, 62(3): 299-322.